

Elevated Ubiquitin Phosphorylation by PINK1 Contributes to Proteasomal Impairment and Promotes Neurodegeneration

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eLife Assessment

The **useful** manuscript presents interesting findings in the field of neurodegenerative diseases by highlighting the dual role of phosphorylated ubiquitin (pUb) in cellular proteostasis and neurotoxicity. However, some claims for discovery are supported by unconvincing and **incomplete** evidence that requires further validation. The poor quality of key immunofluorescent images and questionable quantification analysis raise technical concerns.

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Abstract

Ubiquitin (Ub), a key player of protein turnover, can be phosphorylated by PINK1 kinase to generate S65-phosphorylated ubiquitin (pUb). Elevated pUb levels have been observed in aged human brains and human brains with Parkinson's disease. However, how pUb is involved in neurodegeneration remains elusive. Here we show that elevation of pUb is pervasive in a multitude of neurodegenerative conditions, including Alzheimer's disease, aging, and ischemic injury. In cultured cells, proteasomal inhibition by MG132 leads to sPINK1 accumulation, the cytosolic fragment of PINK1, thus promoting Ub phosphorylation. Elevated pUb impairs proteasomal degradation by disrupting covalent ubiquitin chain elongation and noncovalent proteasome-substrate interaction. Conversely, *pink1* knockout mitigates protein aggregation in both aging and ischemic mouse brains, as well as cells treated with MG132. Using AAV2/9 vector to specifically express sPINK1 in mouse hippocampus neurons, we observed cumulative pUb elevation, accompanied by protein aggregation, proteostasis disturbance, neuronal injury, neuroinflammation, and cognitive

impairment. These sPINK1-induced impairments were reversed by co-expressing Ub/S65A phospho-null Ub mutant but exacerbated by Ub/S65E phospho-mimic mutant. As such, pUb elevation can result from declined proteasomal activity in neurodegenerative conditions, while a constant elevation of pUb actively drives neurodegeneration by further inhibiting proteasomal degradation. Our study reveals a new pathogenic pathway of neurodegeneration, highlighting the pUb-mediated feedforward loop as a promising therapeutic target for pharmaceutical intervention.

Introduction

Neurodegeneration, the hallmark of aging and neurodegenerative diseases, is characterized by irreversible loss of neurons, eventually leading to cognitive and motor impairments [1]. Neurodegeneration is often accompanied by the accumulation of ubiquitinated, potentially neurotoxic proteins, which results from impaired protein degradation machinery, particularly the ubiquitin-proteasome system (UPS) [2–4]. Essential for maintaining cellular proteostasis, the UPS not only eliminates misfolded and toxic proteins but also dynamically regulates neuronal structure and function through controlled protein turnover [5–7]. Substrate proteins are first covalently attached with K48-linked polyubiquitin chains, and the ubiquitin chains non-covalently interact with the ubiquitin-receptors in the proteasome thus targeting the substrate protein for UPS degradation [8–10]. The UPS activity can be compromised through a variety of mechanisms associated with neurodegeneration, such as modification and disassembly of the proteasome, reduced ATP level, direct inhibition by amyloid fibrils, as well as oxidative stress [1–3, 11, 12]. While these factors have been extensively investigated with the goal to identify strategies preventing the decline of proteasomal activity, the dynamic changes of ubiquitin, a central player in the UPS, have been largely overlooked in the context of neurodegeneration.

Ubiquitin itself undergoes a multitude of post-translational modifications that can alter its structure and function [13]. One such modification is phosphorylation at residue S65, installed by the kinase PINK1 (PTEN-induced putative kinase 1) [14]. An elevated level of S65-phosphorylated ubiquitin (pUb) has been observed in aged human brain and human brain with Parkinson's disease [15–17]. Additionally, two forms of PINK1 exist in cells, including the full-length PINK1 localized to the mitochondrial membrane [18, 19], and sPINK1, a cytosolic fragment cleaved from PINK1 [20, 21]. Full-length PINK1 is activated upon mitochondrial damage, leading to the phosphorylation of ubiquitin and Parkin, thereby initiating mitophagy, a neuroprotective process for the selective removal of damaged mitochondria [1, 14, 18, 22]. However, recent studies have indicated that severe and persistent mitochondrial stress can paradoxically cause a failure to induce mitophagy and at the same time, a sustained elevation of pUb levels [23]. Given the prevalence of mitochondrial dysfunction in neurodegenerative diseases [1], pUb accumulation may result from the activation of full-length PINK1 with impaired mitophagy.

Alternatively, elevated pUb levels may arise from inhibited proteasomal degradation. The sPINK1, processed from the full-length PINK1 by mitochondria-residing proteases, is normally rapidly degraded by the proteasome *per* N-end rule [20, 21]. As the UPS function is often impaired under neurodegenerative conditions [1], sPINK1 may accumulate and potentially phosphorylate ubiquitin, contributing to the elevation of pUb level.

While pUb elevation has been proposed as a potential biomarker of neurodegenerative diseases [15, 16], its functional role remains poorly defined. *In vitro* studies suggest that phosphorylation alters ubiquitin's structural dynamics and potentially its function [25–27]. Moreover, PINK1 phosphorylation inhibits ubiquitin chain elongation [28]. Since PINK1 can phosphorylate both monomeric ubiquitin and covalently linked polyubiquitin [30], pUb may

also disrupt the noncovalent interaction between polyubiquitin conjugated to the substrate protein and the proteasome. In addition, it has also been reported that Ub/S65E, a phosphomimic mutant, inhibits protein turnover and reduces the cell viability of yeast when exposed to stresses, including oxidative stress and unfolded protein responses [29]. Thus, pUb may not simply be a biomarker but actively contribute to the neurodegenerative process.

Here we provide compelling evidences that pUb elevation is a driving force in neurodegeneration. We demonstrate that pUb levels increase in various neurodegenerative contexts, including Alzheimer's disease (AD), aging, and ischemic injury, resulting from the accumulation of sPINK1 under the conditions with proteasomal impairment. Importantly, we reveal that elevated pUb further inhibits proteasomal activity, forming a self-amplifying cycle that perpetuates protein aggregation and neuronal damage. By dissecting this mechanism from the molecular level to animal model, we uncover a general pathogenic mechanism that operates across a wide spectrum of neurodegenerative disorders.

Results

Elevated pUb levels are a pervasive feature of neurodegeneration

Elevated pUb levels have been observed in the brains of individuals with Parkinson's disease (PD) [15–17]. In the current study, we extend this observation to Alzheimer's disease (AD), the most prevalent neurodegenerative dementia. We found a marked elevation of both PINK1 and pUb in brain samples from AD patients compared to age- and sex-matched controls (Figure 1A, B; Figure 1—table supplementary 1). This finding was further corroborated in the APP/PS1 mouse model of AD, where increased PINK1 and pUb levels were also detected (Figure 1C, D).

Beyond specific neurodegenerative diseases, pUb elevation also appears to be associated with the aging process itself [15, 16], a physiological condition often accompanied by gradual neurodegeneration. Here we observed a significant increase in neuronal pUb levels in aged wild-type mouse brains compared to young brains (Figure 1E, F). On the other hand, pUb levels in *pink1*-knockout mice remained unchanged with age and were notably lower than in aged wild-type mice (Figure 1E, F).

We further showed that acute neurodegenerative conditions, such as cerebral ischemia, also exhibit elevated pUb levels. In a mouse middle cerebral artery occlusion (MCAO) model, we observed a marked increase in both PINK1 and pUb levels within the ischemic core (Figure 1G, figure 1—figure supplementary 1). To further explore the relationship between pUb and ischemic stress, we subjected HEK293 cells to oxygen-glucose deprivation (OGD), a model mimicking ischemic conditions. Following OGD and reperfusion, we noted a progressive increase in PINK1, sPINK1, and pUb levels, accompanied by a rise in protein aggregation (Figure 1H, I).

Taken together, these findings demonstrate that pUb elevation is a common feature across a wide spectrum of neurodegenerative conditions, both chronic and acute. While this reinforces pUb's potential role as a biomarker, the diverse functions of ubiquitin suggest that elevated pUb levels may actively contribute to the pathogenesis of neurodegeneration.

Ubiquitin phosphorylation by sPINK1 impairs proteasomal function in HEK293 cells

The activation of the full-length PINK1 is triggered by mitochondrial damage, thereby initiating mitophagy [31]. In contrast, sPINK1 is typically degraded rapidly [22], with its accumulation inversely correlating with proteasomal activity. Indeed, we observed that mitochondrial injury by using CCCP and O/A treatment primarily elevate full-length PINK1, while MG132 predominantly

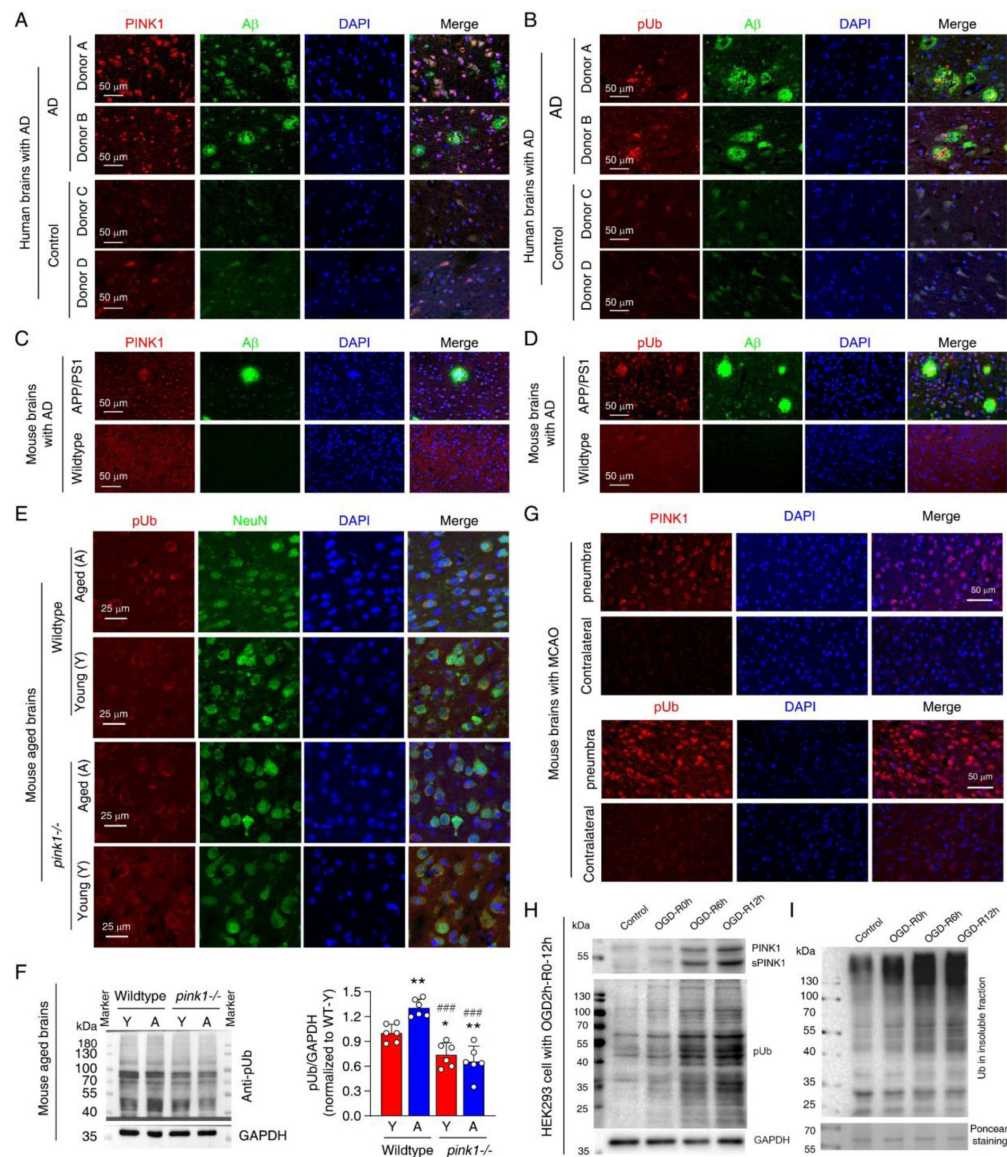


Figure 1.

Elevated pUb levels in neurodegenerative conditions.

(A, B) Double immunofluorescence staining showing the distribution of PINK1 and Aβ in (A), and pUb and Aβ in (B), within the brains of AD patients compared to age-matched controls. Detailed donor demographics are available in Supplementary Table 1.

(C, D) Double immunofluorescence staining of PINK1 and Aβ (C), and pUb and Aβ (D) in the brains of wildtype and APP/PS1 transgenic mice.

(E) Double immunofluorescence staining for pUb and the neuronal marker NeuN in young and aged brains of both wildtype and *pink1*^{-/-} mice.

(F) Western blot analysis of pUb levels in young (Y) and aged (A) wildtype and *pink1*^{-/-} mouse brains, quantitatively comparing protein levels across ages and genotypes. **P*<0.05, ***P*<0.01 for comparisons with young wildtype mice; ###*P*<0.001 for comparisons with aged wildtype mice, one-way ANOVA.

(G) Immunofluorescence staining for PINK1 and pUb in the contralateral and penumbra of mouse brains subjected to MCAO for 2 hours followed by 24 hours of reperfusion. Locations of the analyzed regions are indicated in Figure S1.

(H) Western blot analysis of PINK1 and pUb levels in HEK293 cells subjected to OGD for 2 hours, followed by reperfusion at 0, 6, and 12 hours, to study changes over time.

(I) Western blot analysis of ubiquitin in the insoluble fraction of HEK293 cells post 2-hour OGD and subsequent 0, 6, 12 hours of reperfusion.

increases sPINK1 in HEK293 cells (**Figure 2A** [↗](#)). Further experiments showed that MG132 causes a concentration- and time-dependent increase in sPINK1 and pUb levels, with sPINK1 levels reaching a plateau after 6 hours (**Figures 2B-E** [↗](#)). In contrast, PINK1 and pUb levels remained low in *pink1*-knockout HEK293 cells treated with MG132 (**Figures 2B-E** [↗](#)), paralleling observations from organismal studies (**Figure 1** [↗](#)).

Although PINK1 activation usually involves dimerization at the mitochondrial outer membrane [[19](#) [↗](#), [31](#) [↗](#)], we investigated whether sPINK1 can also phosphorylate ubiquitin in the cytoplasm. We transiently transfected HEK293 cells with various PINK1 constructs: full-length PINK1, a stable cytoplasmic variant of sPINK1 (sPINK1*, PINK1/F101M/L102-L581), a kinase-dead version of this variant (sPINK1*-KD, with K219A/D362A/D384A mutations introduced) [[32](#) [↗](#)], and Ub^{GG}-sPINK1, a short-lived version of native sPINK1 with a ubiquitin appended at the N-terminus [[20](#) [↗](#)]. Western blot analysis confirmed the expression of PINK1 and sPINK1* post-transfection, but not of Ub^{GG}-sPINK1 (**Figure 2F** [↗](#)). Western blot analysis also showed that transfection with both PINK1 and sPINK1* increased pUb levels, with sPINK1* inducing a greater increase than the full-length protein. However, transfection with sPINK1*-KD or Ub^{GG}-sPINK1 did not increase pUb levels notably (**Figure 2F** [↗](#)).

To determine whether the elevation of pUb perturbs protein degradation while minimizing its effect on mitochondria, we transiently transfect the various sPINK1 constructs to increase pUb levels. Immunofluorescence showed diffuse cytoplasmic ubiquitin staining in untransfected or eGFP-transfected control cells, whereas ubiquitin-positive puncta were evident following MG132 treatment or in sPINK1*-transfected cells (**Figure 2G** [↗](#)). Ubiquitin-positive puncta were weakly visible in Ub^{GG}-sPINK1-transfected cells (**Figure 2G** [↗](#), 5th row, indicated with arrows) but absent in sPINK1*-KD-transfected cells. Western blot analysis further confirmed that MG132 treatment and sPINK1* transfection increased ubiquitin levels in the insoluble fraction, while the lack of strong signal with Ub^{GG}-sPINK1 transfection indicates a dose-dependent effect of PINK1 activity (**Figure 2H** [↗](#)).

The activation of PINK1 can also influence autophagy, and ubiquitination is a key step for both autophagic and proteasomal degradation. Thus, the elevation of pUb may increase protein aggregation via the inhibition of both autophagic and UPS degradation. To assess the relative contribution of these two pathways on pUb-promoted protein aggregation, we treated the cells with puromycin to increase ubiquitinated protein level (**Figure 2I** [↗](#), with schematic shown in the left panel). The increase of ubiquitin signal could be markedly reduced by sPINK1* overexpression in comparison to neighboring cells (**Figure 2I** [↗](#)). Blocking the autophagic degradation using BALA, on the other hand, we found that sPINK1*-overexpression increased ubiquitin signal in comparison to neighboring cells (**Figure 2I** [↗](#)). Thus, sPINK1* enhances autophagic degradation of ubiquitinated proteins, possibly via p62 phosphorylation by PINK1 as previously reported [[24](#) [↗](#)], while at the same time, inhibited proteasomal degradation.

To further confirm the inhibitory effect of pUb in proteasomal degradation, we assessed proteasomal activity in cells transfected with Ub-R-GFP, a model substrate for the assessment of cellular proteasomal activity [[33](#) [↗](#), [34](#) [↗](#)]. GFP was rapidly degraded by functional proteasomes but accumulated upon the administration of MG132. On the other hand, transfection with sPINK1* also caused GFP accumulation, while sPINK1*-KD had no effect (**Figure 2J** [↗](#)). This confirms that sPINK1* transfection lead to increased pUb levels, which impairs proteasomal degradation, mirroring the effect of MG132.

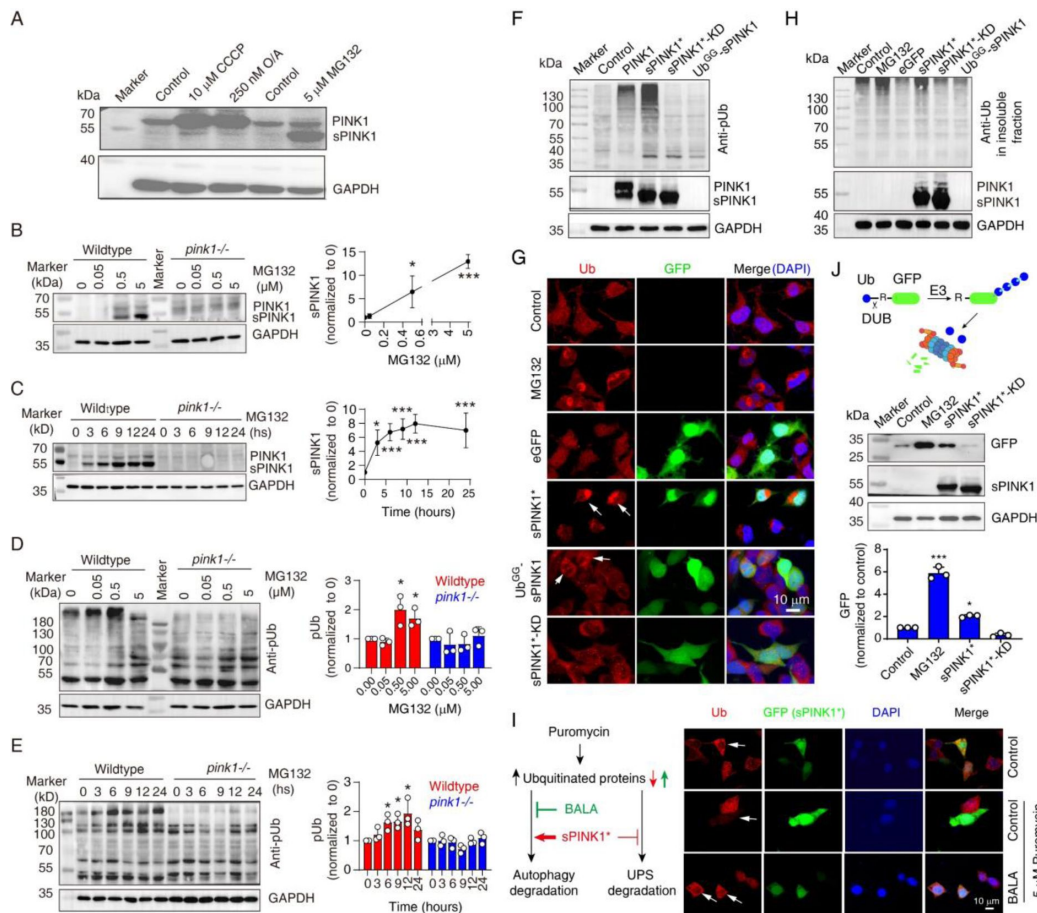


Figure 2.

Impact of sPINK1-mediated Ub phosphorylation on proteasomal activity in HEK293 cells.

- (A) Representative Western blot showing levels of PINK1 following the administration with CCCP, O/A, and MG132.
- (B) Western blot analysis detailing the concentration-dependent effects of MG132 (0-5 μ M) over 8 hours on PINK1 levels. $N=3$; $*P < 0.05$, $***P < 0.001$, compared to 0 μ M MG132, one-way ANOVA.
- (C) Western blot analysis showing the time-dependent effects of 5 μ M MG132 on PINK1 levels over 0-24 hours. $N=4$; $*P < 0.05$, $***P < 0.001$, compared to 0 hours, one-way ANOVA.
- (D) Western blot analysis of pUb levels under a concentration gradient of MG132 (0-5 μ M) for 8 hours. $N=3$; $*P < 0.05$, compared to 0 μ M MG132, one-way ANOVA.
- (E) Western blot analysis of pUb levels over a time course of 24 hours with 5 μ M MG132 treatment. $N=3$; $*P < 0.05$, compared to 0 hours, one-way ANOVA.
- (F) Western blot analysis showing levels of pUb and PINK1 following transfection with various PINK1 constructs: sPINK1* (PINK1/F101M/L102-L581), sPINK1*-KD (kinase-dead sPINK1* with additional K219A/D362A/D384A mutations), and Ub^{GG}-sPINK1 (a short-lived version of native sPINK1 with an appended N-terminal Ub).
- (G) Representative immunofluorescence images of Ub staining in cells treated with MG132 or transfected with various sPINK1 constructs, highlighting differences in ubiquitin localization and aggregation. The white arrows indicate positively transfected cells.
- (H) Western blot analysis of Ub in the insoluble fraction of cell lysates following treatment with 5 μ M MG132 or after transfection with different sPINK1 constructs.
- (I) Immunofluorescence staining of Ub differentiating the roles of sPINK1 over-expression on proteasomal and autophagic degradation of puromycin increased unfolded proteins. The left panel illustrates the experimental strategy. Puromycin (5 μ g/ml) was applied 2 hours before harvesting with or without 0.1 μ M BALA (an autophagy inhibitor). The white arrows indicate positively transfected cells.
- (J) Western blot analysis showing GFP degradation in HEK293 cells, with the top panel illustrating the Ub-GFP construct, a model substrate to evaluate proteasomal degradation efficiency. $N=3$; $*P < 0.05$, $***P < 0.001$, compared to the control group, one-way ANOVA.

pUb impairs proteasomal activity via covalent and noncovalent mechanisms

To understand how ubiquitin phosphorylation affects UPS degradation, we first assessed ubiquitin chain formation *in vitro*. While unmodified ubiquitin efficiently formed di-, tri-, and higher-order chains with E1 and E2 enzymes, pUb predominantly produced di-Ub chains (**Figure 3A** [↗](#)), consistent with the previous report [28 [↗](#)]. The introduction of S65A (phospho-null) and S65E (phospho-mimetic) mutations into ubiquitin provided further insights into the effects of phosphorylation. The Ub/S65A mutation facilitated chain elongation similar to the unmodified Ub, whereas Ub/S65E mutation largely resulted in di-Ub formation but not higher-order chains (**Figure 3A** [↗](#)). Although the Ub/S65E mutant may not completely mimic all characteristics of pUb, it effectively demonstrates the impact of S65 phosphorylation on chain formation.

We also investigated K48-linked ubiquitin chain formation in HEK293 cells expressing FLAG-tagged Ub/K48-only mutant (FLAG-Ub/48K). Following proteasome inhibition with MG132, K48-linked chains accumulated in wildtype and *pink1*^{-/-} cells, but not in cells overexpressing full-length PINK1 or sPINK1* (**Figure 3B** [↗](#)).

Since PINK1 can phosphorylate both ubiquitin monomers and polyUb chains [30 [↗](#)], we next investigated the impact of pUb chains on proteasomal degradation. *In vitro* degradation assays with purified proteasomes showed that degradation of K48-linked pUb-chain-modified GFP (pK48-polyUb-GFP) was slower compared to the unmodified counterpart (K48-polyUb-GFP) (**Figure 3C** [↗](#), figure 3—figure supplementary 1, 2). The interactions between ubiquitinated substrates and immobilized proteasomes were then analyzed using total internal reflection fluorescence (TIRF) microscopy following an established protocol [35 [↗](#)]. The assays revealed fewer Ub-GFP puncta for pK48-polyUb-GFP compared to unphosphorylated controls (**Figure 3D**, **E** [↗](#), figure 3—figure supplementary 3), and these puncta dissociated more rapidly from the proteasomes, suggesting a shorter dwell time of the proteasome-substrate complex (**Figure 3F-I** [↗](#), figure 3—figure supplementary 4).

Collectively, these findings demonstrate that elevated pUb levels not only disrupt covalent ubiquitin chain elongation and subsequent substrate ubiquitination, but also hinder the non-covalent binding of ubiquitinated substrates to the proteasome. Both mechanisms significantly contribute to the impairment of proteasomal activity.

pink1 knockout mitigates protein aggregation resulting from declined proteasomal activity

Aging and neurodegenerative diseases commonly feature accumulations of protein aggregates alongside decreased proteasomal activity [36 [↗](#)]. Our findings have shown that conditions with impairing proteasomal function, such as Alzheimer's disease, cerebral ischemia, aging, and pharmacologic treatment of proteasome inhibitor, result in increased levels of sPINK1 and pUb. As elevated pUb could in turn inhibit the UPS degradation, we hypothesized that proteasomal inhibition and pUb elevation form a positive feedback loop contributing to protein aggregation, while *pink1* knockout disrupts this process (**Figure 4A** [↗](#)).

In support of this hypothesis, we found that aged wildtype mice exhibited distinct Ub-positive puncta in neurons and increased insoluble ubiquitin levels, indicative of protein aggregation. In contrast, aged *pink1*^{-/-} mice showed no such age-related changes in ubiquitin staining or insoluble ubiquitin levels (**Figure 4B**, **C** [↗](#)). Following MCAO, insoluble ubiquitin levels were significantly elevated in the ipsilateral hemisphere of wildtype mice, but not in *pink1*^{-/-} mice (**Figure 4D** [↗](#)).

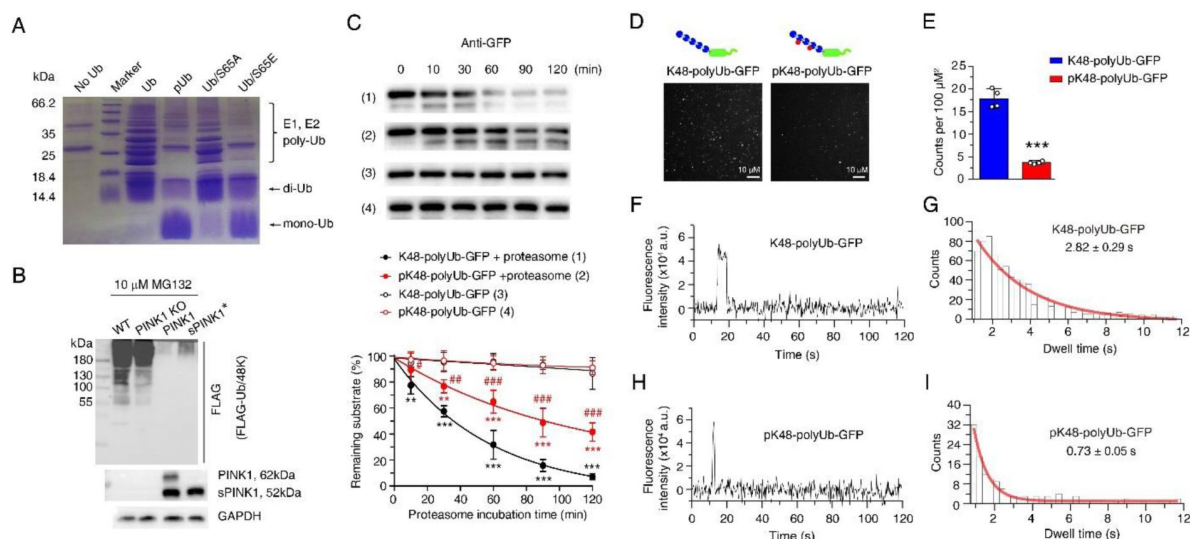


Figure 3.

Impact of ubiquitin phosphorylation on chain elongation and proteasomal association.

(A) Coomassie blue staining showing Ub chain formation using different Ub variants as building blocks: wildtype Ub, phosphorylated Ub (pUb), phospho-null Ub/S65A, and phospho-mimic Ub/S65E.

(B) Analysis the transfection and knockout of PINK1 on K48-linked Ub chain formation in HEK293 cells treated with 10 μ M MG132. Cells were transfected with FLAG-tagged Ub/48K. Additionally, *pink1*^{-/-} cells were transfected with either full-length PINK1 or the sPINK1* variant.

(C) *In vitro* proteasomal degradation assay of GFP modified with K48-linked ubiquitin chains (K48-polyUb-GFP) versus with phosphorylated K48-linked ubiquitin chains (pK48-polyUb-GFP). $N=3$; ** $P<0.01$, *** $P<0.001$ compared to respective controls without added proteasome; # $P<0.05$, ## $P<0.01$, ### $P<0.001$ compared to K48-polyUb-GFP with proteasome, two-way ANOVA.

(D) Representative TIRF microscopy images showing the binding of K48-polyUb-GFP (left) and pK48-polyUb-GFP (right) to surface-immobilized proteasomes, visualized as bright puncta.

(E) Quantitative analysis of puncta density from TIRF images, with $N=4$; *** $P<0.001$ compared to K48-polyUb-GFP, using a paired *t*-test.

(F, H) Representative fluorescence traces of a single punctum for K48-polyUb-GFP (F) and pK48-polyUb-GFP (H), detailing the dynamic interactions at the proteasome surface.

(G, I) Analysis of GFP fluorescence dwell time for K48-polyUb-GFP (G) and pK48-polyUb-GFP (I). Data are binned and modeled with a single exponential decay curve (red line), illustrating differences in substrate-proteasome interaction durations.

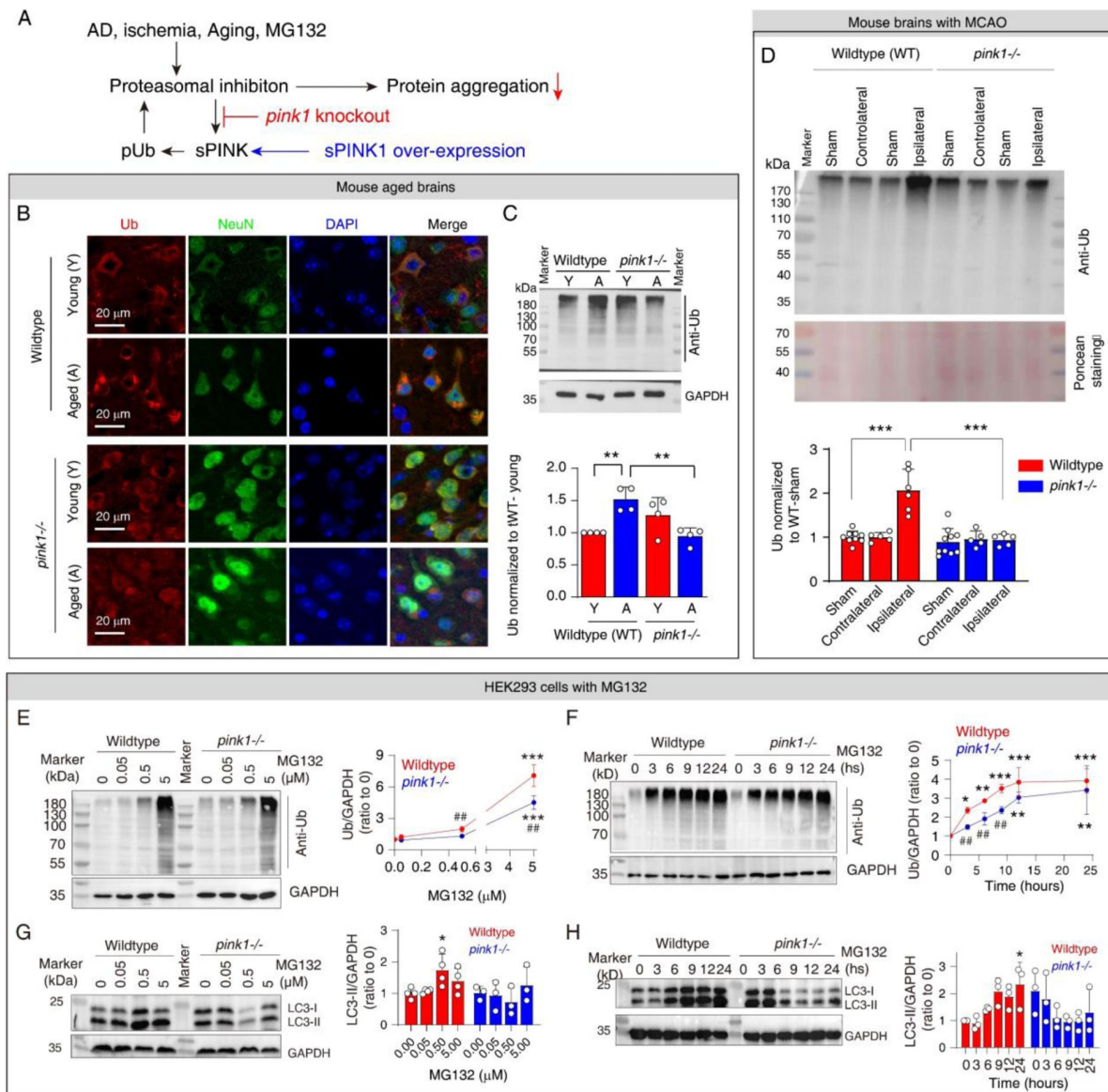


Figure 4.

pink1 knockout mitigates protein aggregation under proteasomal inhibition.

(A) Schematic representation illustrating the reciprocal feed-forward cycle among proteasomal inhibition, and the elevation of sPINK1 and pUb levels.

(B) Immunofluorescence staining of Ub in the brains of young and aged wildtype and *pink1*^{-/-} mice.

(C) Western blot analysis quantifying Ub in the insoluble protein fraction from young and aged mouse brains. Statistical analysis includes $N=4$; $^{**}P<0.01$, one-way ANOVA.

(D) Western blot analysis of Ub in the insoluble protein fraction of wildtype and *pink1*^{-/-} mouse brains subjected to MCAO for 2 hours followed by 24 hours of reperfusion. Comparison includes the sham-operated group (same procedure without occlusion). Locations of the analyzed regions (contralateral and ipsilateral) are indicated in Figure 1—figure supplementary 1.

(E) Western blot analysis of Ub levels in the insoluble fraction following treatment with 0-5 μ M MG132 for 8 hours. $N=3$; $^{***}P<0.001$ compared to 0 μ M MG132; $^{##}P<0.01$ compared with wildtype cells, one-way ANOVA.

(F) Western blot analysis of Ub levels in the insoluble fraction following treatment with 5 μ M MG132 over a 0-24-hour period. $N=4$; $^{*}P<0.05$, $^{**}P<0.01$, $^{***}P<0.001$ compared to 0 μ M MG132; $^{##}P<0.01$ compared with wildtype cells, one-way ANOVA.

(G) Western blot analysis of LC3 levels following treatment with 0-5 μ M MG132 for 8 hours. $N=4$; $^{*}P<0.05$ compared to 0 μ M MG132, one-way ANOVA.

(H) Western blot analysis of LC3 levels after 5 μ M MG132 treatment over 0-24 hours. $N=3$; $^{*}P<0.05$ compared to 0 μ M MG132, one-way ANOVA.

To directly assess the link between proteasomal inhibition, elevated pUb level, and protein aggregation, we treated wildtype and *pink1*^{-/-} cells with MG132. As expected, MG132 induced an increase in insoluble ubiquitin levels in both cell types in a concentration- and time-dependent manner (**Figure 4E, F**). However, the accumulation of ubiquitin-positive aggregates occurred significantly slower in *pink1*^{-/-} cells, even though they eventually reached levels comparable to those in wildtype cells at 24 hrs after MG132 administration (**Figure 4F**). This suggests that PINK1 deficiency delays the onset and progression of protein aggregation, but does not completely prevent it. Intriguingly, MG132 treatment also increased LC3-II levels, a marker of autophagy, in wildtype cells, possibly indicating either activation of autophagy or reduced LC3 degradation as a compensatory response to proteasomal inhibition (**Figure 4G, H**). Conversely, LC3-II levels did not increase in *pink1*^{-/-} cells following MG132 treatment (**Figure 4G, H**). This indicates that PINK1 is required for autophagy activation under proteasome stress, aligning with our observations of enhanced autophagy upon PINK1 overexpression (**Figure 2I**).

Together, the results indicate that *pink1* knockout can alleviate protein aggregation associated with aging, ischemic injury, or proteasomal inhibition by disrupting the self-amplifying cycle of proteasomal impairment (**Figure 4A**).

sPINK1* overexpression causes cumulative proteome-wide changes

As illustrated in **Figure 4A**, the over-expression of sPINK1 may lead to a sustained elevation of pUb and cause protein aggregation. Since protein aggregation is one of primary causes for neurodegeneration, we assessed the impact of pUb elevation on neurons. First, we confirmed the roles of sPINK1 and phosphomimetic Ub/S65E mutant on protein aggregation in SH-SY5Y cell, a neuroblast-link cell line. Ubiquitin-positive puncta was observed in cells treated with MG132 and overexpressing either sPINK1* or Ub/S65E (Figure 5—figure supplementary 1), similar to the findings in HEK293 cell (**Figure 2G**).

It has been shown that the phospho-null mutant Ub/S65A could mitigate, while the phospho-mimic Ub/S65E could mimic, the effects produced by PINK1 through the phosphorylation of ubiquitin [29]. Thus, we specifically overexpressed sPINK1*, sPINK1* with Ub/S65A (sPINK1*+Ub/S65A), Ub/S65E, or GFP control in mouse hippocampal neurons using AAV2/9, and the specific expression was confirmed through GFP co-localization with fluorescence staining of NeuN in the CA1 region (Figure 5—figure supplementary 2). While no changes were observed at 10 days post-transfection, significant neuronal loss was evident at 30 and 70 days in mice expressing Ub/S65E, but not in other three groups (Figure 5—figure supplementary 2). Additionally, Ub/S65E induced glial responses at 30- and 70-days post transfection, as demonstrated by GFAP/Iba1 staining and GFAP/CD11b Western blots (Figure 5—figure supplementary 3, 4). In contrast, the sPINK1* and sPINK1*+Ub/S65A groups showed an increase in GFAP (indicative of astrocyte activation) at 70-days post transfection, without effects on Iba1 or CD11b (Figure 5—figure supplementary 3, 4). The existence of GFAP response may be caused by potential neuronal injury at 70-days post-transfection.

To investigate the molecular underpinnings of sPINK1*-induced neuronal injury, we performed proteomic analysis of mouse hippocampus at 30- and 70-days post-transfection (Figure 5—dataset supplementary 1). Despite similar number of total proteins identified between the GFP and sPINK1* groups (Figure 5—figure supplementary 5A-D), notable differences emerged. For example, the neuronal injury/neuroinflammation marker HMGB1 was elevated at both time points with sPINK1* overexpression, while the mitochondrial marker Tom20 showed a decrease only at 70 days, indicating a delayed loss of mitochondria.

Gene set enrichment analysis (GSEA) further revealed a time-dependent impact on mitochondrial function. At 30-days, changes in proteins related to mitophagy, mitochondrial organization, and respiratory electron transport suggested an initial compensatory response to mitochondrial stress (Figure 5—figure supplementary 5D). However, at 70-days, these changes expanded to include broader mitochondrial dysfunction and apoptotic processes, culminating in mitochondrial loss and neuronal injury (Figure 5—figure supplementary 5D).

Further GO analysis underscored the extensive proteomic dysregulation caused by sPINK1* overexpression (Figure 5—figure supplementary 5F-H). At 70-days, a large number of proteins were both upregulated and downregulated, indicative of disrupted proteostasis. This could be attributed to proteasomal inhibition or a stress response to protein aggregates. For example, we observed a significant up-regulation of RNA-binding and RNA-processing proteins. Similar proteome-wide change has been noticed in the Ub/S65E over-expressed yeast [29]. Importantly, we found many affected proteins were associated with the disruption of synaptic functions and neuronal projections, while downregulated proteins were linked to myelination, synaptic maturation, and lipid metabolism (Figure 5—figure supplementary 5F-H). Note that many of these changes were insignificant at 30-days post-transfection (Figure 5—figure supplementary 5F-H). These findings collectively underscore the systemic impact of sPINK1* overexpression on neuronal structure and function, which likely drives progressive neuronal injury.

Elevated pUb levels drive protein aggregation in mouse hippocampal neurons

To assess the impact of elevated pUb on protein aggregation *in vivo*, we examined PINK1 and pUb levels in the mouse hippocampus following sPINK1* overexpression. At 30-days, a distinct band for sPINK1 was observed, accompanied by a significant increase in pUb levels (Figure 5A). At 70-days, three distinct bands that could be assigned to sPINK1, full-length PINK1, and ubiquitinated PINK1 (Figure 5B). The presence of full-length PINK1 indicates mitochondrial injury, while ubiquitinated PINK1 suggests compromised proteasomal activity. Notably, co-expression of Ub/S65A yielded only a faint sPINK1 band (Figure 5B), indicating its potential to counteract the proteasomal inhibition induced by sPINK1*, thereby promoting degradation of this stable sPINK1 variant. Consequently, pUb levels were elevated with sPINK1* overexpression but remained at control levels with sPINK1* and Ub/S65A co-expression (Figure 5C).

Immunofluorescence staining revealed ubiquitin puncta in hippocampal neurons overexpressing sPINK1*, sPINK1*+Ub/S65A, or Ub/S65E (Figure 5D). Western blot analysis showed an increase in soluble ubiquitin in brains from the sPINK1*+Ub/S65A group, likely arising from the overexpressed Ub/S65A (Figure 5E). Levels of insoluble ubiquitin were significantly elevated in the sPINK1*, sPINK1*+Ub/S65A, and Ub/S65E groups (Figure 5F). Notably, proteins smaller than 70 kDa accumulated in the sPINK1* and Ub/S65E groups, indicating the inhibition of ubiquitin chain elongation as observed *in vitro* (Figure 3B), but not in the sPINK1*+Ub/S65A group. Furthermore, the overexpression of sPINK1*, sPINK1*+Ub/S65A, or Ub/S65E had minimal impact on autophagic flux, as indicated by stable LC3-II and p62 levels (Figure 5—figure supplementary 6). These findings suggest that the observed protein aggregation is primarily driven by elevated pUb levels and the resulting proteasomal dysfunction.

Elevated pUb levels impair neuronal integrity and cognitive function

To determine the impact of pUb-induced protein aggregates on neuronal function, we conducted behavioral tests on mice at 70-days post-transfection. Overexpression of sPINK1* or Ub/S65E significantly impaired cognitive functions, evidenced by reduced performance in novel object

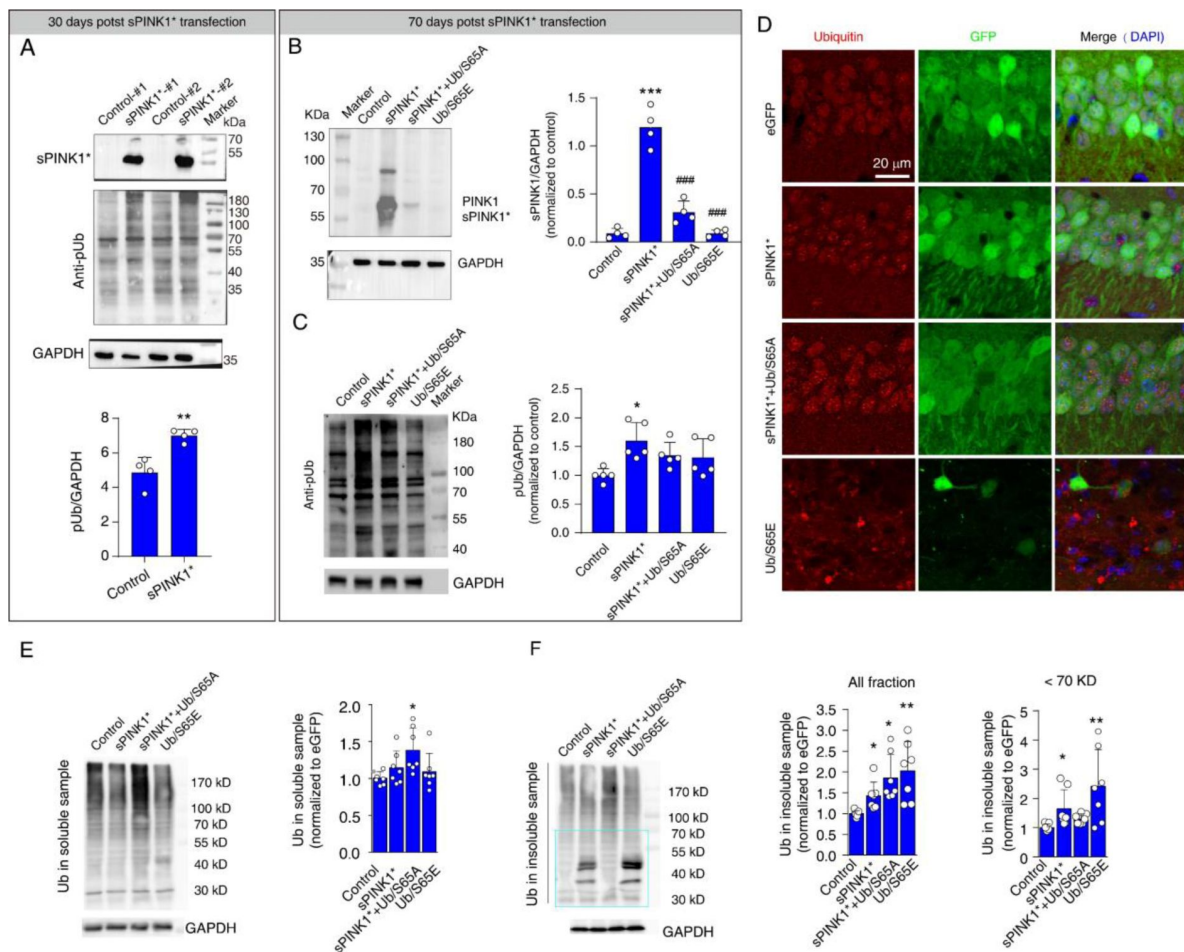


Figure 5.

Elevated pUb levels induce protein aggregation in mouse hippocampal neurons.

(A) Western blot analysis showing levels of PINK1 and pUb in mouse hippocampus at 30-days post-transfection. $N=4$; $**P<0.01$ compared with control, one-way ANOVA.

(B) Western blot analysis of PINK1 in the mouse hippocampus at 70-days post-transfection, illustrating changes in expression over time. $N=4$; $**P<0.001$ compared with control; $###P<0.001$ compared with sPINK1, one-way ANOVA.

(C) Western blot analysis of pUb in the mouse hippocampus at 70-days post-transfection, highlighting the persistence of phosphorylation changes. $N=5$; $*P<0.05$ compared with control, one-way ANOVA.

(D) Representative immunofluorescence images depicting ubiquitin staining in the CA1 region of the mouse hippocampus at 70-days post-transfection.

(E) Western blot analysis of ubiquitin in the soluble fraction of hippocampal lysates at 70-days post-transfection. $N=7$; $*P<0.05$ compared with control, one-way ANOVA.

(F) Western blot analysis of Ub in the insoluble fraction of hippocampal lysates at 70-days post-transfection. Quantitative analysis includes total protein and specifically proteins smaller than 70 kDa. $N=7$; $*P<0.05$, $**P<0.001$ compared with control, one-way ANOVA.

recognition (**Figure 6A, B**) and fear conditioning tests (**Figure 6C, D**). Notably, co-expression of Ub/S65A with sPINK1* ameliorated sPINK1*-induced cognitive deficits, suggesting a protective effect against pUb-caused impairments.

Consistent with these behavioral findings, Western blot analysis showed decreased levels of Tom20, a mitochondrial marker, in brains overexpressing sPINK1*. This mitochondrial deficit was reversed when Ub/S65A was co-expressed (**Figure 6E**), aligning with the proteomic data (Figure 5—figure supplementary 5D). Additionally, we noted reduced levels of MAP2, a dendritic marker, and PSD95, a postsynaptic density marker, in the brains expressing sPINK1* and Ub/S65E. These reductions were restored to normal levels upon Ub/S65A co-expression (**Figure 6F, G**). Accordingly, Golgi staining revealed a decrease in dendritic spine density following sPINK1* overexpression, which was reversed upon Ub/S65A co-expression (**Figure 6H**). In contrast, Ub/S65E overexpression led to more severe dendritic and spine loss, with notably slimmer dendrites and a significant reduction in spine density compared to sPINK1* (**Figure 6H**).

These findings underscore that sPINK1*-induced neuronal damage accumulates over time and can be effectively mitigated by co-expressing the phospho-null mutant Ub/S65A. Conversely, the phospho-mimic Ub/S65E exacerbates neuronal damage rapidly. Collectively, these results demonstrate that elevated pUb level contributes to hippocampal neuronal injury and associated cognitive impairments, highlighting its significance as a driver of neurodegeneration.

Discussion

In the current study, we demonstrate that elevated pUb level is a pervasive feature in both chronic and acute neurodegenerative conditions. This observation aligns with previous reports of sustained pUb elevation in human brains with PD [15, 16]. Our findings extend this phenomenon to AD, aging, and cerebral ischemia. While severe mitochondrial injury has been implicated as a mechanism leading to pUb elevation [23, 37], we have uncovered an additional contributing factor. We show that inhibition of proteasomal degradation stabilizes sPINK1 in the cytoplasm, where it retains its ability to phosphorylate ubiquitin.

Proteasomal inhibition can be caused by various pathological factors in neurodegeneration [1]. Cerebral ischemia, for instance, induces ATP deficiency, while hypoxia and the secondary effect of oxidative stress would further lead to proteasomal modification and impairment of proteasomal function [11, 38]. On the other hand, in AD, amyloid fibrils can directly block the proteasome's substrate channel [2, 12]. Due to the decline of proteasomal activity in various neurodegenerative diseases, elevated sPINK1 levels would consequently result in the cumulative increase of pUb. Thus, the elevation of ubiquitin phosphorylation also constitutes a hallmark of neurodegeneration.

We further demonstrate that elevation of pUb level reciprocally contributes to UPS dysfunction. Our findings reveal that increasing pUb levels through sPINK1* overexpression exacerbated protein aggregation by impairing proteasomal degradation in both *in vitro* and *in vivo* models. Conversely, *pink1* knockout reduces protein aggregation in aged mouse brains, brains subjected to MCAO, and HEK293 cells treated with a proteasome inhibitor. Mechanistically, we have found that ubiquitin phosphorylation negatively impacts both the covalent elongation of ubiquitin chains, as previously reported [28], and the non-covalent interaction between K48-linked polyUb chains and the proteasome, as discovered in the current study, likely resulting from the increased ubiquitin structural dynamics upon phosphorylation [26, 39]. This dual inhibitory mechanism hampers ubiquitination, recruitment, and ultimately, degradation of substrate proteins by the proteasome. Impaired proteasomal activity further impedes sPINK1 degradation,

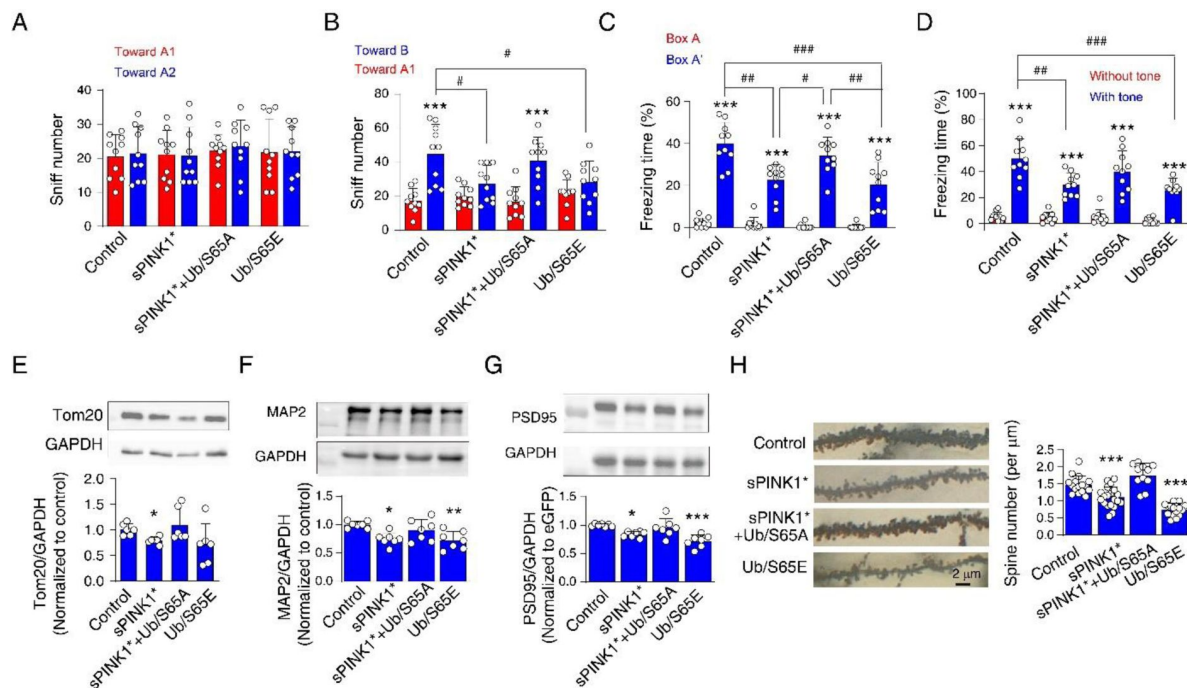


Figure 6.

Elevated pUb levels induce neuronal injury in mouse brains.

(A, B) Novel object recognition test showing the sniff number to objects during training (A) and testing (B) phases. $N=10$;

*** $P<0.001$ compared with the sniff number toward A1 object during training, paired t -test. # $P<0.05$, one-way ANOVA.

(C, D) Fear conditioning tests evaluating contextual (C) and cued (D) memory. $N=10$; *** $P<0.001$ compared with Box A (contextual) or absence of tone (cued), paired t -test. # $P<0.05$, ## $P<0.01$, ### $P<0.001$, one-way ANOVA.

(E-G) Western blot analysis measuring levels of mitochondrial, dendritic, and synaptic markers: Tom20 (E), MAP2 (F), and PSD95 (G) in the mouse hippocampus. $N=7$; * $P<0.05$, ** $P<0.01$, *** $P<0.001$ compared with control, one-way ANOVA.

(H) Golgi staining used to assess hippocampal neuronal spines and dendrites. The left panel shows representative images; the right panel provides a statistical analysis of spine density on hippocampal neuron dendrites from 3 mice. *** $P<0.001$ compared with control, one-way ANOVA.

forming a self-amplifying cycle of elevated pUb levels and progressive proteasomal dysfunction, culminating in the accumulation of protein aggregates, the established hallmark of neurodegeneration.

Our findings indicate that elevated pUb levels can actively contribute to neurodegeneration. Yet, the functional role of pUb appears nuanced: while transient pUb elevation is generally neuroprotective, sustained elevation becomes neurotoxic. As a neuroprotective mechanism, PINK1 activation upon mitochondrial damage leads to transient phosphorylation of ubiquitin and Parkin, triggering mitophagy to remove damaged mitochondria [30, 40, 41]. Indeed, PINK1 deficiency has been shown to cause neurodegeneration, prompting the development of small-molecule activators [23, 40, 42]. However, in the current study, overexpression of a stable sPINK1 variant led to chronic neuronal injury, resulting from the reciprocal relationship between proteasomal inhibition and cumulative pUb elevation. The progressive decline in proteasomal function would consequently disrupt cellular proteostasis and allow protein aggregates to build up. As such, pUb is not simply a biomarker for neurodegeneration — sustained elevation is a driving factor for progressive neuronal injury and neuroinflammation.

To confirm the neurotoxic effect of pUb, we utilized the phospho-null Ub/S65A mutant to dilute pUb and antagonize its effects. Ub/S65A restored proteasomal activity *in vivo* and even promoted the degradation of the stable sPINK1 variant, preventing its accumulation. Moreover, Ub/S65A co-expression allowed the formation of longer polyUb chains, further supporting its protective role. Conversely, overexpression of the phospho-mimetic Ub/S65E mutant induced even more severe neuronal damage than sPINK1* overexpression. This effect can be attributed to the 100% increase in mimicking pUb with Ub/S65E, whereas sPINK1*-induced pUb levels are low and subjected to dephosphorylation [43, 44]. Similar toxic effects resulting from Ub/S65E overexpression has been reported in yeast [29]. Furthermore, it should be noted that, sPINK1* can phosphorylate both free and conjugated Ub, whereas Ub/S65E only mimics monomeric pUb, potentially further contributing to the detrimental effects of Ub/S65E mutant.

In conclusion, our findings have unveiled a novel pathogenetic mechanism of neurodegeneration, in which the self-amplifying cycle of elevated pUb level and proteasomal inhibition leads to the progressive decline of proteasomal activity and the accumulation of toxic protein aggregates. Our findings demonstrate pUb is more than a consequence of this gradual cumulative process, but also a critical contributor to disease progression. Targeting this pUb-mediated self-amplifying cycle can be a viable therapeutic strategy to ameliorate the progression of neurodegenerative diseases. On the other hand, elevated pUb levels observed in a wide spectrum of neurodegenerative diseases may hinder the pharmacological efficacy of drugs that target the UPS.

Materials and Methods

Human brain immunofluorescence staining

The human brain cingulate gyrus paraffin section samples include two samples from people with Alzheimer's Disease and two age-matched samples as the control. The samples with AD were obtained from the Netherlands Brain Bank (NBB), Netherlands Institute for Neuroscience, Amsterdam (Project number 1613), and the age matched control samples were obtained from the National Health and Disease Human Brain Tissue Resource Center, Hangzhou, China (Project number CN20240454). All material has been collected from donors for or from whom a written informed consent for a brain autopsy and the use of the material and clinical information for research purposes had been obtained by the NBB and CBB. The detailed information of the donors is provided in Supplementary Table 1. The experiments were conducted with approval from the Ethics Committee of the School of Medicine, Zhejiang University, with the approval number ZJU2023-011.

For immunofluorescence staining, the samples were incubated at 62°C overnight to dewax. The samples were then hydrated—Xylene for 3 times, 100% ethanol for 2 times, 95% ethanol for 2 times, 75% ethanol for 2 times, and ddH₂O for 2 times. The samples were placed in EDTA antigen retrieval solution, heated to 95–100°C for 30 minutes, and cooled down at room temperature for 30 minutes. After washing with PBS buffer for 3 times, the samples were placed in 3% H₂O₂ solution for 10 minutes. After washing with PBS buffer for 3 times, the samples were incubated in 0.3% Triton-X PBS for 30 minutes. After washing with PBS buffer for 3 more times, the samples were permeabilized with 0.3% Tween in PBS for 30 minutes, and rinsed 3 times with PBS to remove the detergent. The samples were incubated in a solution with 5% Sudan black B (Aladdin Biochemical Technology Co., Ltd, S109070-25g, Shanghai, China) and 70% ethanol for 1 min at room temperature.

Following PBS rinse, the samples were covered with the blocking solution (5% normal goat serum, 2% bovine serum albumin, 0.1% Tween prepared in TBS-T) for 1 hr. The samples were incubated with rabbit anti-PINK1 antibody (1:200, Novus, Colorado, USA, BC100-494), rabbit anti-pUb antibody (1:200, Millipore, Massachusetts, USA, ABS1513), or mouse anti-A β antibody (1:500, Biologend, California, USA, SIG-39320) at 4 °C overnight. After washing with PBS for 3 times, the samples were incubated with Cy3-conjugated anti-Rabbit IgG (1:200, Jackson, Pennsylvania, USA, 711-165-152) and fluorescein (FITC)-conjugated anti-mouse IgG (1:200, Jackson, Pennsylvania, USA, 715-096-150) for 2 hrs at room temperature. After washing with PBS for 3 times, the samples were mounted on the slides using a ProLong™ Gold Antifade Mountant with DAPI (Invitrogen Corp., Carlsbad, CA, USA). The immunofluorescence images were taken using an Olympus FV100 confocal microscope.

Animals

The C57BL/6J and APP/PS1 mice were purchased from Zhejiang Academy of Medical Science. All mice were kept in the Laboratory Animal Center, Zhejiang University School of Medicine. The mice had free access to water and food in air-conditioned rooms (~26 °C, relative humidity ~50%) on a 12-h light/dark cycle. Mice were handled following the Guide for the Care and Use of the Laboratory Animals of the National Institutes of Health. The experimental protocols were approved by the Ethics Committee of Laboratory Animal Care and Welfare, Zhejiang University School of Medicine, with the approval number ZJU20190138.

Using CRISPR-Cas9-mediated genome editing technology [45], the *pink1* gene knockout mice (C57BL/6J) were customized purchased in Transgenic Mouse Laboratory of Laboratory Animal Center of Zhejiang University. Briefly, two sgRNA sequences that target exon 6 of the *pink1* gene were cloned into pX330-U6-Chimeric_BB-CBH-hSpCas9 plasmid (a gift from Feng Zhang (Addgene plasmid # 42230; <http://n2t.net/addgene>; 42230; RRID: Addgene 42230) using the *BbsI* restriction site. The sequence of sgRNA1 is 5'-GACAGCCATCTGCAGAGAGG-3', and the sequence of sgRNA2 is 5'-GCAGGCAGGACTCACCTCAG-3'. The knockout of *pink1* gene was confirmed by using DAN genotype, quantitative PCR, and Western blotting.

Mouse hippocampus CA1 injection of Adeno-associated virus (AAV)

The rAAV-EF1a-WPRE-hGH pA (AAV2/9) was used to specifically express proteins within neurons, a bistrionic vector expressing GFP and sPINK1* (PINK1/F101M/L102-L581) as two separate proteins. Moreover, the rAAV-EF1a-Ub/S65A-P2A-sPINK1-P2A-EGFP-WPRE-hGH pA can express GFP, sPINK1*, and Ub/S65A as three separate proteins, and rAAV-EF1a-Ub/S65E-P2A-EGFP-WPRE-hGH pA can express GFP and Ub/S65E as two separate proteins.

Mice of 9-months old were anesthetized with pentobarbital sodium (80 mg/kg) by intraperitoneal injection (i.p.). Then the mice were fixed on a mouse stereotaxic holder (RWD Life Science Co. LTP, China) for the injection of AAV. The distinct AAV was stereotactically injected bilaterally into the CA1 region using a microinjector (Hamilton syringe) with a 31-gauge needle and micropump.

(LEGATO 130 syringe pump, KD Scientific). After anesthesia using pentobarbital sodium (100 mg/kg, i.p.), the injections were made at stereotaxic coordinates of Bregma: anteroposterior (AP) = -1.7 mm, mediolateral (ML) = 1.5 mm, dorsoventral (DV) = 1.7 mm. A total 0.3 μ l AAV (2.25×10^{12} viral particles/ml) was injected into each side of the hippocampus.

Behavioral tests

Novel object recognition (NOR) was used to evaluate the spatial memory of mice according to the previous report [46]. After handling for 5 days, the test was performed on the 68th day post injection. An open-field test system (ViewPoint Behavior Technology, France) was used for NOR detection. During the training, two identical objects (named object A1 and A2, 5 cm' 5 cm' 5 cm blue cone) was placed diagonally. The mouse was placed at the center of the box, and the movement of the mouse was recorded for 10 min. After returning to the home cage for 24 hrs, the mouse was put in the same box with one of the objects replaced by a novel object in distinct color and shape (object B, 5 cm' 5 cm' 10 cm yellow cuboid). The sniff number (number of visits) toward the objects was analyzed.

The fear conditioning test was performed in the Fear Conditioning System (Coulbourn Instruments, USA) according to the previous report [46]. The freezing percentage was recorded during each phase to analyze their memory. During the training phase, the mouse was placed in the box (identified as box A) without any stimulation for 1.5 minutes. Then, a tone (3000 Hz, 85 db) was applied for 30 seconds, and an electric shock (1 mA) was applied at the final 2 seconds. After the shock, the mouse was kept in box A for 30 second, and was put back to the home cage. The testing was performed at 24 hrs after training. The mouse was placed back in the box A without any stimulation for 5 minutes, while the freezing percentage in box A indicates contextual memory. Two hours after the test in box A, the mouse was placed in a new box (identified to box B). For the first 2 minutes, no stimulus was given, and subsequently, the same tone (3000 Hz, 85 db) was applied for the following 3 minutes. The freezing percentage in box B without and with tone was used to analyze the cue memory of the mice.

Focal cerebral ischemia/reperfusion and Nissl staining

After anesthetized, the mice were operated with middle cerebral artery occlusion (MCAO) for 2 hrs ischemia followed by 24 hrs reperfusion. Briefly, a 6-0 nylon monofilament suture was inserted into the internal carotid to occlude the origin of MCA [47]. Two hours after occlusion, the suture was withdrawn to allow reperfusion for 24 hrs. Mouse body temperature was maintained at 37 °C using a thermostatic pad during and after surgery. In sham-operated mice, the external carotid artery was surgically isolated but the suture was no inserted.

For Nissl staining, the brain slices were incubated in a mixture of acetone and chloroform (1:1) for 15 min, and then sequentially incubated in 100%, 95%, 70% alcohol for 5 min. Then the slices were stained for 10 min in Nissl staining containing 0.2% purple crystal (Yuanhang Reagent Factory, Shanghai, China, YHSJ-01-92) and 0.3% acetic acid. Then, the slices were dehydrated by incubation in 70%, 95%, 100% alcohol. Finally, the slices were put into xylene for 5 min and then mounted with a mixture of neutral resin and xylene (1:1). The images were taken under a microscope (Olympus BX51, Japan).

Mouse brain tissue collection

Mice were anesthetized using pentobarbital sodium (150 mg/kg, i.p.). The mouse brains were removed after transcardially perfused with 4 °C saline and freshly prepared 4% paraformaldehyde. The brains were further fixed in 4% paraformaldehyde at 4 °C for one day and transferred to 30% sucrose solution for dehydration. The brains were sliced into 30 μ m thick slices

by cryomicrotomy (CM1900, Leica, Wetzlar, Germany). The slices were stored at -20°C in the solution (30% glycerol, 30% glycol, 40% PBS) for immunofluorescence staining, or stored in 2.5% glutaraldehyde for 12 hrs at 4°C for transmission electron microscopy (TEM) analysis.

For Western blotting, the mouse was transcardially perfused only with 4°C saline. The brains were removed, and the hippocampus was isolated. The hippocampus was quickly frozen in liquid nitrogen and stored at -80°C until use.

Cell lines, cell culture, and cell transfection

HEK293 cells and SH-SY5Y cells were purchased from the Institute of Cell Biology of the Chinese Academy of Sciences (Cell Biology of the Chinese Academy of Sciences, Shanghai, China). The *pink1* gene knockout HEK293 cell was homely made using CRISPR-Cas9-mediated genome editing technology[48]. PINK1 sgRNA (CCTCATCGAGGAAAAACAGG) was cloned to U6-sgPINK1-mCherry plasmid (a gift from John Doench & David Root (Addgene plasmid # 78038; <http://n2t.net/addgene:78038>; RRID:Addgene_78038)). The U6-sgPINK1-mCherry plasmid and PX458-Cas9-EGFP plasmid (a gift from Prof. Feng Zhang (Broad Institute, Cambridge, MA), Addgene plasmid #48138) were co-transfected into HEK293 cells. The mCherry and GFP double-positive cells were single-cell sorted 24 h post-transfection using an MoFloAstrios EQ cell sorter (Beckman Coulter, US) and grown in separate cultures that were subsequently screened for the presence of frameshift mutations leading to nonsense-mediated decay on both alleles. PINK1 KO was confirmed using western blotting.

Cells were grown in Dulbecco's modified essential medium (DMEM, Gibco by Thermo Fisher Scientific, C11995500BT) supplemented with 10% heat-inactivated fetal bovine serum (FBS, Zhejiang Tianhang Biotechnology, China, 11011-8611) in the atmosphere of 5% CO_2 at 37°C .

Lipo3000 (Invitrogen, California, USA, L3000001) was used to transfect plasmids, following to the manufacturer' protocol. Briefly, an appropriate number of cells were seeded one day before transfection with 70-80% confluent at the time of transfection. Before transfection, plasmid DNA was carefully mixed with 250 μl Opti-MEM (Gibco by Life Technologies, Carlsbad, USA). Subsequently, 250 μl Lipo3000 was carefully mixed with Opti-MEM. The mixture was incubated at room temperature for 25 min. The freshly formed DNA/Lipo3000 precipitates were carefully pipetted to the cells. The medium containing transfection reagents was removed after 6 hrs and fresh medium was added.

At the end of treatments, the cells on cover slides were washed with 37°C PBS and then fixed using freshly prepared 4% paraformaldehyde. Subsequently, the cells were stored in PBS under 4°C for immunofluorescence staining.

Oxygen glucose deprivation and reperfusion (OGD/R)

Cells were washed twice by using glucose-free DMEM (Procell, PM150270) and incubated at 37°C with the ventilation of 95% N_2 and 5% CO_2 for 2 hours. The control cells were washed twice by using high glucose DMEM, and incubated in 95% air and 5% CO_2 under 37°C for 2 hours. Then the cells were cultured in the high glucose DMEM with 10% FBS in 95% air and 5% CO_2 under 37°C . This was identified as reperfusion.

The construction of Ub-R-GFP vector

The Ub-R-GFP is used for the *in vivo* measurement of proteasomal degradation activity, and constructed according to a previous report[34]. Briefly, the ubiquitin open reading frame was amplified by PCR with the following primers:

Sense primer: 5'-GCG GAATTCACCATGCAGATCTTCGTGAAGACT-3'

Antisense primer: 5'-GCG GGATCCTGTCGACCAAGCTTCCCGCGCCACCTCTGAGACGGAGTAC-3'

The PCR product was cloned into the EcoRI and BamHI sites of the EGFP-N2 vector. Beside the R residue, a 12 amino acid peptide was inserted between Ub and GFP to increase the proteasomal degradation. Thus, the final product is Ub-R-GKLGRQDPPVAT-GFP. The accumulation of GFP protein was used to determine the proteasomal degradation activity.

Immunofluorescence staining

The floating brain slices or cells on the cover glasses were incubated in 0.1% Triton-X PBS for 30 min, and followed by the incubation in 5% donkey serum for 1 hr. The samples were then incubated with mouse anti-ubiquitin antibody (1:200, Santa Cruz Biotechnology, Texas, USA; SC-8017), rabbit anti-pUb antibody (1:200, Millipore, Massachusetts, USA; ABS1513), rabbit anti-PINK1 antibody (1:200, Novus, Colorado, USA; BC100-494), mouse anti-GFAP antibody (1:1000, CST, Pennsylvania, USA; #3670), rabbit anti-Iba1 antibody (1:1000, FUJIFILM Wako Pure Chemical Corporation, Tokyo, Japan; 019-19741), rabbit anti-NeuN antibody (1:1000, CST, Pennsylvania, USA; #36662), mouse anti-A β (1:500, Biolegend, California, USA, SIG-39320) at 4 °C overnight. After washed with PBS (10 min' 3 times), the samples were incubated with Cy3 AffiniPure Donkey Anti-Mouse IgG (H+L) (1:200, Jackson ImmunoResearch, PA, USA; 715-605-150) or Cy3 AffiniPure Goat Anti-Rabbit IgG (H+L) (1:200, Jackson ImmunoResearch, PA, USA; 715-605-150; 111-165-003) or Alexa Fluor488 AffiniPure Goat Anti-Rabbit IgG (H+L) (1:200, Jackson ImmunoResearch, PA, USA; 111-545-003) or Alexa Fluor488 AffiniPure Goat Anti-Rabbit IgG (H+L) (1:200, Jackson ImmunoResearch, PA, USA; 715-545-150) for 2 h. After washing with PBS (10 min each for 3 times), the slices were mounted on slides using a ProLong Gold Antifade Mountant with DAPI. The images were taken under an Olympus FV100 confocal microscope (Olympus, Japan) and analyzed using MetaMorph (version 7.8.0.0, Molecular Devices, LLC. San Jose, CA 95134 USA).

Western blot analysis

The cells and brain tissues for the Western blot were lysed using RIPA lysis buffer (Beyotime Biotechnology Research Institute, Jiangsu, China; P0013B) with protease inhibitor (Beyotime Biotechnology Research Institute, Jiangsu, China; P1005) and phosphatase inhibitor (Beyotime Biotechnology Research Institute, Jiangsu, China; P1081). The cells were lysed on ice for 30 min, and every 10 min, the cells were gently vortexed. The brain samples were thoroughly homogenized with a precooled TissuePrep instrument (TP-24, Gering Instrument Company, Tianjin, China) for 1 min at 4 °C. Subsequently, the lysate was centrifuged at 12,000 rpm for 30 min at 4 °C. The supernatant was first collected as the soluble protein sample for Western blot analysis. The precipitate was resuspended in 20 μ l SDS buffer (2% SDS, 50 mM Tris-HCl, pH7.5) for ultrasonic pyrolysis at 4 °C. The ultrasonic pyrolysis cycle included 10-sec ultrasonic pyrolysis (Diagenode, Seraing, Belgium) and 30-sec interval, for a total of 8 cycles. The samples were then centrifuged at 12,000 rpm for 30 min at 4 °C. The supernatant was again collected as the insoluble protein sample for Western blot analysis.

For brain tissues, the mice were anesthetized by intraperitoneal injection of pentobarbital sodium (150 mg/kg) before sacrifice. The brains were quickly removed after transcardially perfused with 4 °C saline. The hippocampus was separated and stored at -80 °C until usage.

All protein concentration was determined by using BCA Protein Assay Kit (Beyotime Biotechnology, Shanghai, China; P0009). Appropriate protein samples (50-100 μ g) were used for Western blot analysis. The following antibodies were used: mouse anti-Ub antibody (1:800, Santa Cruz Biotechnology, Texas, USA; SC-8017), rabbit anti-pUb antibody (1:1000, Millipore, Massachusetts, USA; ABS1513), rabbit anti-PINK1 antibody (1:1000, Novus, Colorado, USA; BC100-494), mouse anti-GFAP antibody (1:1000, CST, Pennsylvania, USA; #3670), rabbit anti-CD11b antibody (1:1000, Abcam, Cambridge, UK; ab133357), mouse anti-GAPDH antibody (1:5000, Proteintech, Wuhan, China; 60004-1-Ig), mouse anti-MAP2 antibody (1:2000, Millipore, Billerica,

MA, USA; AB5622), rabbit anti-PSD95 antibody (1:1000, CST, Pennsylvania, USA; 3450S), rabbit anti-Tom20 antibody (1:1000, Pennsylvania, USA; 42406S), rabbit anti-LC3 antibody (1:1000, Sigma, Massachusetts, USA; L7543), rabbit anti-p62 antibody (1:1000, Abcam, California, USA; ab109012), mouse anti-FLAG antibody (1:10000, TransGen biotech, Beijing, China; HT201-01). The secondary antibody was HRP-conjugated goat anti-mouse IgG (1:3000, Cell Signaling Technology, MA, USA; 7076S) or HRP-conjugated goat anti-rabbit IgG (1:10000, Jackson ImmunoResearch, PA, USA, 111-035-003).

The immunoblots were then detected using ECL reagents (Potent ECL kit, MultiSciences Biotech, Hangzhou, China; P1425) and measured using GBOX (LI-COR, Odyssey-SA-GBOX, NE, USA). The results were normalized to GAPDH or Ponceau staining (Beyotime Biotechnology Research Institute, Jiangsu, China; P0022), or to the eGFP-expression control on the same immunoblot membrane.

Golgi staining

On the 70th-day post-transfection, mice were anesthetized by intraperitoneal injection of pentobarbital sodium (150 mg/kg) before sacrifice. The mouse brain was quickly removed, and the dorsal hippocampus was separated. The hippocampus was immediately fixed in the fixative (Servicebio, Wuhan, China, G1101) for over 48 hrs.

The dorsal hippocampus was sliced with a thickness of 2-3 mm thickness around the injection site. The tissue was gently rinsed with PBS at least 3 times. The hippocampus tissue was then placed in Golgi-cox staining solution (Servicebio, Wuhan, China; G1069), and incubated in a dark for 14 days. The Golgi-cox staining solution was changed 48 hrs after the first soak, and every 3 days afterwards. On the 14th-day after staining, the tissues were immersed in distilled water for 3 times and then incubated in 80% acetic acid overnight until the tissue became soft. After rinsing with distilled water, the tissue was placed into 30% sucrose.

The tissue was cut into 100 μ m slices with an oscillating microtome, and the slices were placed on a gelatin slide and dried overnight in the dark. The sliced were then treated with ammonia water for 15 min. After washing with distilled water, the slices were incubated in the fixing solution for 15 min. After washing, the slices were sealed using glycerin gelatin. Images were taken using VS120 Virtual Slide Microscope (Olympus, Japan).

The *in vitro* ubiquitin-dependent proteasome degradation

Ubiquitin was prepared as previously described [49, 50]. K48-linked di-ubiquitin and tetra-ubiquitin were prepared following an established protocol [51], with the conjugation reaction catalyzed by 2.5 μ M human E1 and 20 μ M E2-25K, in 20 mM pH 8.0 Tris-HCl buffer. Ub/K48R mutant was incorporated as the distant Ub, and Ub/77D mutant was incorporated as the proximal Ub to prepare Ub chain of the desired length. The residue D77 at the C-terminus was removed with hydrolase YUH1. The fusion protein His-TEV-Ub-GFP was prepared recombinantly based on the design previously described [34, 52]. The *Pediculus humanus corporis* PINK1 kinase (pHPINK1) was prepared and purified as previously described [26, 27]. Ub phosphorylation was confirmed with electrospray mass spectrometry (Agilent G6530 Q-TOF).

The 26S proteasome from HEK293 cells was prepared following an established protocol, with the purification tag appended at the C-terminus of Rpn11 [53, 54]. The proteasome activity was assessed with a fluorogenic peptide, N-succinyl-Leu-Leu-Val-Tyr-7-amido-4-methyl coumarin (Sigma, Massachusetts, USA; S6510). A final concentration of 100 nM proteasome and 100 μ M proteasome peptide substrate was prepared in the reaction buffer, containing 50 mM Tris-HCl pH 7.5, 4 mM ATP (Sigma-Aldrich, Cat# A6559) and 5 mM MgCl₂. The fluorescence intensity of the product was measured by a fluorometer (Horiba Scientific, FluoroMax-4) at an excitation wavelength of 360 nm.

For Western blot analysis of GFP, 200 nM proteasomal substrate and 10 nM human 26S proteasome were prepared in 20 mM Tris pH 8.0 buffer, with 50 mM NaCl, 5 mM ATP, and 5 mM MgCl₂. The mixture was incubated at 37 °C for 0, 10, 30, 60, 90, and 120 min. At each time point, 20 µL samples were taken for Western blot analysis. A mouse anti-GFP antibody (1:2000, San-Ying Proteintech Group, 66002-1) and an HRP-conjugated goat anti-mouse IgG (1:3000, CST, Pennsylvania, USA; 7076S) were used. ImageJ was used to analyze the band intensities, which are normalized to the initial intensity.

The recombinant Ub, pUb (phosphorylated by phPINK1), Ub/S65A, and Ub/S65E were reacted with 2.5 µM human E1 and 20 µM E2-25K in 20 mM pH 8.0 Tris-HCl buffer. The formed Ub chain was determined by Coomassie blue staining on SDS page gel.

TIRF analysis of ubiquitin-proteasome interaction

The coverslips were prepared following an established protocol [55]. Each coverslip was divided into multiple lanes with double-sided tape for parallel TIRF experiments, thus to ensure repeatability with different proteins added to the same slide. Streptavidin (VWR Life Science, Cat# 97062-808) in imaging buffer at a concentration of 50 µg/ml was loaded; excess unbound streptavidin was washed away. The imaging buffer contains 25 mM Tris-HCl (pH 7.5), 50 mM NaCl, 10 mM MgCl₂, 40 mM imidazole, 5 mM BSA (Sigma-Aldrich), 2.5 mM ADP (Sigma-Aldrich), and 0.5 mM ATP-γ-S (Sigma-Aldrich), as previously described [35].

The purified human 26S proteasome at 20 nM in the imaging buffer was added to the streptavidin-immobilized coverslip, with excess unbound proteins washed away with the imaging buffer. The substrate protein, either K48-polyUb-GFP or pK48-polyUb-GFP, was then added to the coverslip to a final concentration of 400 pM. The imaging was performed on a Nikon A1 TIRF microscope equipped with a 488 nm laser and an EMCCD camera (Andor DU-897). The time series were acquired at 300 ms per frame for a total duration of 2 min for each time series. The view area was 512 pixels by 512 pixels (16 by 16 µm), and the experiments were repeated four times on four different proteasome-immobilized cover slides. The puncta density was counted in iSMS software [56] with default settings. Only when the substrate protein becomes bound to the immobilized proteasome, the green puncta can be observed, and only the puncta with brightness above a certain threshold (≥ 3000 a.u.) and the dwell time of more than three consecutive frames are selected for counting the overall number of puncta. The dwell times were tabulated in 300 ms bins and fitted with single exponential decay.

Proteomics analysis

Hippocampus samples were rinsed using PBS and lysed in RIPA lysis buffer. The samples were then denatured at 95 °C for 5 min, followed by sonication at 4 °C (3 s on, 3 s off, with 30% amplitude). The lysate was centrifuged at 16,000 g for 10 min at 4 °C, and the supernatant was collected as whole tissue extract. Protein concentration was determined by Bradford protein assay. Extracts from the protein sample (2 g) were trypsin digested.

The enriched peptides sample were precipitated in solution A (0.1% formic acid), and was centrifuged at 16,000 g for 10 min. The peptides were separated on a reverse-phase nano-HPLC C18 column (Precolumn, 3 µm, 120 Å, 2×100 i.d.; analysis column, 1.9 µm 120 Å, 30 cm x150 µm, i.d.) at a flow rate of 600 nL/min with a 150 min gradient of 7-95% solution B (0.1% formic acid in acetonitrile). For peptide ionization, 2100 V was applied, and a 320 °C capillary temperature was used. For detection with Q Exactive HF, peptides were analyzed with one full scan (350–1400 m/z, R = 120,000 at 200 m/z) with an automatic gain control target of 3×10^6 ions, with max injection time of 80 ms, followed by up to 30 data-dependent MS/MS scans with high-energy collision dissociation (target 5×10^4 ions, max injection time 19 ms, isolation window 1.6 m/z, normalized collision energy of 27%), detected in the Orbitrap (R = 15,000 at 200 m/z). Dynamic exclusion time was set as 30 sec.

Raw MS data were searched against the mouse National Center for Biotechnology Information (NCBI) Refseq protein database (updated on 2013/07/01, 29764 entries) by the software Thermo Proteome Discoverer 2.1 using Mascot 2.3. The mass tolerances were 20 ppm for precursor and 50 mmu for product ions for Q Exactive HF. The search engine set Acetyl(Protein N-term), oxidation(M), Phospho(ST), Phospho(Y) as variable modifications. Trypsin digestion of up to two missed cleavages was allowed. The peptide identifications were accepted at a false discovery rate (FDR) of 1%. Using a label-free approach, a unique peptide was used to represent the absolute abundance of a particular peptide across the sample.

For the proteomic data analysis, quantile normalization was applied to eliminate batch effect, and missing values were imputed using the minimum value observed in the dataset. The differentially expressed proteins were defined with the following these criteria: an increase by more than two-fold or decrease by more than 50%. For Gene Set Enrichment Analysis (GSEA) analysis, Gene Ontology (GO) analysis was conducted by GSEAPy (version 0.10.5) Python package to evaluate the enriched GO terms. The association of proteins with GO terms was derived from the DAVID database. For Gene Set Enrichment Analysis (GSEA), the ranking metric used was the fold change, and significantly enriched GO terms were filtered with a cutoff *P*-value of 0.05.

Statistical analysis

Data are presented as mean $\pm$ SD. GraphPad Prism Software (version 6.0, GraphPad Software, San Diego, CA, USA) was used for statistical analysis and plot graphs. The ROUT test was used to identify outliers. The Brown-Forsythe test was used to evaluate the equal variances of the data. D'Agostino-Pearson omnibus normality test was performed to test the normality of the data. If the data pass the normality test and equal variance test, *t*-test, the parametric one-way ANOVA (Tukey multiple comparisons test), or two-way ANOVA (Newman-Keuls multiple comparisons test) was used. If not, nonparametric test (Dunn's multiple comparisons test) was used. Chi-square test was also used to assess the differences in categorical data distributions. $P < 0.05$ was considered statistically significant.

Author contributions

W.P.Z. and C.T. designed the study and wrote the manuscript. C.C., T.Y.G., H.W.Y., Y.Z., T.Y., T.W., and T.F.W. conducted the experiments. Y.B.L. and W.P.Z. analyzed the data at mouse and cellular level. T.T.L. analyzed the proteomic data. C.T. analyzed the data at protein level.

Acknowledgements

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Data availability

All data needed to evaluate the conclusions in the paper are present in the paper and/or the Supplementary Materials.

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Reviewer #1 (Public review):

Summary:

The manuscript discusses the role of phosphorylated ubiquitin (pUb) by PINK1 kinase in neurodegenerative diseases. It reveals that elevated levels of pUb are observed in aged human brains and those affected by Parkinson's disease (PD), as well as in Alzheimer's disease (AD), aging, and ischemic injury. The study shows that increased pUb impairs proteasomal degradation, leading to protein aggregation and neurodegeneration. The authors also demonstrate that PINK1 knockout can mitigate protein aggregation in aging and ischemic mouse brains, as well as in cells treated with a proteasome inhibitor. While this study provided some interesting data, several important points should be addressed before being further considered.

Strengths:

- (1) Reveals a novel pathological mechanism of neurodegeneration mediated by pUb, providing a new perspective on understanding neurodegenerative diseases.
- (2) The study covers not only a single disease model but also various neurodegenerative diseases such as Alzheimer's disease, aging, and ischemic injury, enhancing the breadth and applicability of the research findings.

Weaknesses:

- (1) PINK1 has been reported as a kinase capable of phosphorylating Ubiquitin, hence the expected outcome of increased p-Ub levels upon PINK1 overexpression. Figures 5E-F do not demonstrate a significant increase in Ub levels upon overexpression of PINK1 alone, whereas the evident increase in Ub expression upon overexpression of S65A is apparent. Therefore,

the notion that increased Ub phosphorylation leads to protein aggregation in mouse hippocampal neurons is not yet convincingly supported.

(2) The specificity of PINK1 and p-Ub antibodies requires further validation, as a series of literature indicate that the expression of the PINK1 protein is relatively low and difficult to detect under physiological conditions.

(3) In Figure 6, relying solely on Western blot staining and golgi staining under high magnification is insufficient to prove the impact of PINK1 overexpression on neuronal integrity and cognitive function. The authors should supplement their findings with immunostaining results for MAP2 or NeuN to demonstrate whether neuronal cells are affected.

(4) The authors should provide more detailed figure captions to facilitate the understanding of the results depicted in the figures.

(5) While the study proposes that pUb promotes neurodegeneration by affecting proteasomal function, the specific molecular mechanisms and signaling pathways remain to be elucidated.

<https://doi.org/10.7554/eLife.103945.1.sa2>

Reviewer #2 (Public review):

Summary:

The manuscript makes the claim that pUb is elevated in a number of degenerative conditions including Alzheimer's Disease and cerebral ischaemia. Some of this is based on antibody staining which is poorly controlled and difficult to accept at this point. They confirm previous results that a cytosolic form of PINK1 accumulates following proteasome inhibition and that this can be active. Accumulation of pUb is proposed to interfere with proteostasis through inhibition of the proteasome. Much of the data relies on over-expression and there is little support for this reflecting physiological mechanisms.

Weaknesses:

The manuscript is poorly written. I appreciate this may be difficult in a non-native tongue, but felt that many of the problems are organisational. Less data of higher quality, better controls and incision would be preferable. Overall the referencing of past work is lamentable. Methods are also very poor and difficult to follow.

Until technical issues are addressed I think this would represent an unreliable contribution to the field.

<https://doi.org/10.7554/eLife.103945.1.sa1>

Reviewer #3 (Public review):

Summary:

This study aims to explore the role of phosphorylated ubiquitin (pUb) in proteostasis and its impact on neurodegeneration. By employing a combination of molecular, cellular, and in vivo approaches, the authors demonstrate that elevated pUb levels contribute to both protective and neurotoxic effects, depending on the context. The research integrates proteasomal inhibition, mitochondrial dysfunction, and protein aggregation, providing new insights into the pathology of neurodegenerative diseases.

Strengths:

- The integration of proteomics, molecular biology, and animal models provides comprehensive insights.
- The use of phospho-null and phospho-mimetic ubiquitin mutants elegantly demonstrates the dual effects of pUb.
- Data on behavioral changes and cognitive impairments establish a clear link between cellular mechanisms and functional outcomes.

Weaknesses:

- While the study discusses the reciprocal relationship between proteasomal inhibition and pUb elevation, causality remains partially inferred.
- The role of alternative pathways, such as autophagy, in compensating for proteasomal dysfunction is underexplored.
- The immunofluorescence images in Figure 1A-D lack clarity and transparency. It is not clear whether the images represent human brain tissue, mouse brain tissue, or cultured cells. Additionally, the DAPI staining is not well-defined, making it difficult to discern cell nuclei or staging. To address these issues, lower-magnification images that clearly show the brain region should be provided, along with improved DAPI staining for better visualization. Furthermore, the Results section and Figure legends should explicitly indicate which brain region is being presented. These concerns raise questions about the reliability of the reported pUb levels in AD, which is a critical aspect of the study's findings.
- Figure 4B should also indicate which brain region is being presented.

<https://doi.org/10.7554/eLife.103945.1.sa0>

Author response:

Public Reviews:

Reviewer #1 (Public review):

Summary:

The manuscript discusses the role of phosphorylated ubiquitin (pUb) by PINK1 kinase in neurodegenerative diseases. It reveals that elevated levels of pUb are observed in aged human brains and those affected by Parkinson's disease (PD), as well as in Alzheimer's disease (AD), aging, and ischemic injury. The study shows that increased pUb impairs proteasomal degradation, leading to protein aggregation and neurodegeneration. The authors also demonstrate that PINK1 knockout can mitigate protein aggregation in aging and ischemic mouse brains, as well as in cells treated with a proteasome inhibitor. While this study provided some interesting data, several important points should be addressed before being further considered.

Strengths:

- (1) Reveals a novel pathological mechanism of neurodegeneration mediated by pUb, providing a new perspective on understanding neurodegenerative diseases.*
- (2) The study covers not only a single disease model but also various neurodegenerative diseases such as Alzheimer's disease, aging, and ischemic injury, enhancing the breadth and applicability of the research findings.*

Weaknesses:

- (1) PINK1 has been reported as a kinase capable of phosphorylating Ubiquitin, hence the expected outcome of increased p-Ub levels upon PINK1 overexpression. Figures 5E-F do*

not demonstrate a significant increase in Ub levels upon overexpression of PINK1 alone, whereas the evident increase in Ub expression upon overexpression of S65A is apparent. Therefore, the notion that increased Ub phosphorylation leads to protein aggregation in mouse hippocampal neurons is not yet convincingly supported.

Indeed, overexpression of sPINK1* alone caused little change in Ub levels in the soluble fraction (Figure 5E), which is expected. Ub in the soluble fraction is in a relatively stable, buffered state. However, overexpression of sPINK1* resulted in an increase in Ub levels in the insoluble fraction, indicating protein aggregation. The molecular weight of Ub in the insoluble fraction was predominantly below 70 kDa, implying that phosphorylation inhibits Ub chain elongation.

To further examine this, we used the Ub/S65A mutant to antagonize Ub phosphorylation, and found that the aggregation at low molecular weight was significantly reduced, indicating a partial restoration of proteasomal activity. The increase in Ub levels in both the soluble and insoluble fractions likely results from the high rate of ubiquitination driven by the elevated levels of Ub. Notably, the overexpressed Ub/S65A was detected in the Western blot using the wild-type Ub antibody, which accounts for the apparently increased Ub level.

When overexpressing Ub/S65E, we again saw an increase in Ub levels in the insoluble fraction (but no increase in the soluble fraction), with low molecular weight bands even more prominent than those observed with sPINK1* transfection. These findings collectively support the conclusion that sPINK1* promotes protein aggregation through Ub phosphorylation.

(2) The specificity of PINK1 and p-Ub antibodies requires further validation, as a series of literature indicate that the expression of the PINK1 protein is relatively low and difficult to detect under physiological conditions.

We acknowledge the challenges in achieving optimal specificity for commercially available and custom-generated antibodies targeting PINK1 and pUb, particularly given the low endogenous levels of these proteins under physiological conditions. Despite these limitations, we observed robust immunofluorescent staining for PINK1 (Figures 1A, 1C, and 1G) and pUb (Figures 1B, 1D, and 1G) in human brain samples from Alzheimer's disease (AD) patients, as well as in mouse brains from models of AD and cerebral ischemia. The significant elevation of PINK1 and pUb under these pathological conditions likely accounts for the clear visualization. To validate antibody specificity, we have included images from *pink1*^{-/-} mice as negative controls in the revised manuscript (Figure 1C and 1D, third panel).

In addition, we detected a significant increase in pUb levels in aged mouse brains compared to young ones (Figures 1E and 1F). Notably, in *pink1*^{-/-} mice, pUb levels remained unchanged between young and aged groups, despite some background signal, further supporting the conclusion that pUb accumulation during aging is PINK1-dependent.

In HEK293 cells, *pink1*^{-/-} cells served as a negative control for PINK1 (Figure 2B and 2C) and for pUb (Figure 2D and 2E). While the Western blot using the pUb antibody displayed some nonspecific background, pUb levels in *pink1*^{-/-} cells remained unchanged across all MG132 treatment conditions (Figures 2D and 2E), further attesting the reliability of our findings.

(3) In Figure 6, relying solely on Western blot staining and Golgi staining under high magnification is insufficient to prove the impact of PINK1 overexpression on neuronal integrity and cognitive function. The authors should supplement their findings with immunostaining results for MAP2 or NeuN to demonstrate whether neuronal cells are affected.

Thank you for raising this important point. We included NeuN immunofluorescent staining in Figure 5—figure supplement 2 of the original manuscript. The results demonstrate a significant loss of NeuN-positive cells in the hippocampus following Ub/S65E overexpression, while no apparent change in NeuN-positive cells was observed with sPINK1* transfection alone. These findings provide evidence of neuronal loss in response to Ub/S65E, further supporting the impact of pUb elevation on neuronal integrity.

While we did not perform MAP2 immunostaining, we included complementary analyses to assess neuronal integrity. Specifically, we performed Western blotting to determine MAP2 protein levels and used Golgi staining to study neuronal morphology and synaptic structure in greater detail. These analyses revealed that overexpression of sPINK1* or Ub/S65E decreased MAP2 levels and caused damage to synaptic structures (Figures 6F and 6H). Importantly, the deleterious effects of sPINK1* overexpression could be rescued by co-expression of Ub/S65A, further underscoring the role of pUb in mediating these changes.

Together, our NeuN immunostaining, MAP2 analysis, and Golgi staining provide strong evidence for the impact of PINK1 overexpression and pUb elevation on neuronal integrity and synaptic health. We believe these complementary approaches sufficiently address the reviewer's concern and highlight the pathological consequences of elevated pUb levels.

(4) The authors should provide more detailed figure captions to facilitate the understanding of the results depicted in the figures.

Figure captions will be updated with more details in the revised manuscript.

(5) While the study proposes that pUb promotes neurodegeneration by affecting proteasomal function, the specific molecular mechanisms and signaling pathways remain to be elucidated.

The specific molecular mechanisms and signaling pathways through which pUb promotes neurodegeneration are likely multifaceted and interconnected. Mitochondrial dysfunction appears to be a central contributor to neurodegeneration following sPINK1* overexpression. This is supported by (1) an observed increase in full-length PINK1, indicative of impaired mitochondrial quality control, and (2) proteomic data revealing enhanced mitophagy at 30 days post-transfection and substantial mitochondrial injury by 70 days post-transfection. The progressive damage to mitochondria caused by protein aggregates can cause further neuronal injury and degeneration.

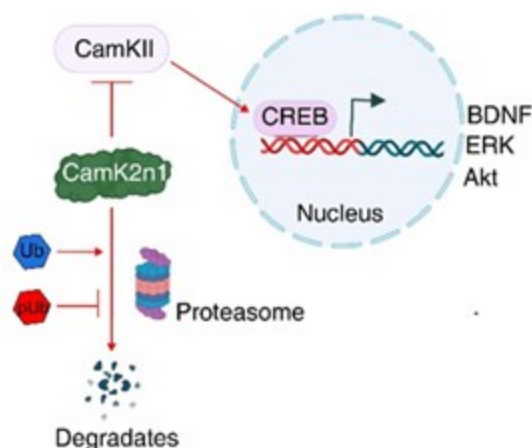
In addition, reduced proteasomal activity may result in the accumulation of inhibitory proteins that are normally degraded by the ubiquitin-proteasome system. Our proteomics analysis identified a >54-fold increase in CamK2n1 (UniProt ID: Q6QWF9), an endogenous inhibitor of CaMKII activation, following sPINK1* overexpression. This is particularly significant because the accumulation of CamK2n1 could suppress CaMKII activation and, subsequently, inhibit the CREB signaling pathway (illustrated below). As CREB is essential for synaptic plasticity and neuronal survival, its inhibition may further amplify neurodegenerative processes.

While our study identifies proteasomal dysfunction and mitochondrial damage as key initial triggers, downstream effects—such as disruptions in signaling pathways like CaMKII-CREB—likely contribute to a broader cascade of pathological events. These findings highlight the complexity of pUb-mediated neurodegeneration and suggest that further exploration of downstream mechanisms is necessary to fully elucidate the pathways involved.

We plan to include the proteomics data, in the revised manuscript, of mouse brain tissues at 30 days and 70 days post-transfection, to further highlight this downstream effect upon

proteasomal dysfunction.

Author response image 1.



Reviewer #2 (Public review):

Summary:

The manuscript makes the claim that pUb is elevated in a number of degenerative conditions including Alzheimer's Disease and cerebral ischemia. Some of this is based on antibody staining which is poorly controlled and difficult to accept at this point. They confirm previous results that a cytosolic form of PINK1 accumulates following proteasome inhibition and that this can be active. Accumulation of pUb is proposed to interfere with proteostasis through inhibition of the proteasome. Much of the data relies on over-expression and there is little support for this reflecting physiological mechanisms.

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Methods are also very poor and difficult to follow.

Until technical issues are addressed I think this would represent an unreliable contribution to the field.

(1) Antibody specificity and detection under pathological conditions

We acknowledge the limitations of commercially available antibodies for detecting PINK1 and pUb. Despite these challenges, our findings demonstrate a significant increase in PINK1 and pUb levels under pathological conditions, such as Alzheimer's disease (AD) and ischemia. Additionally, we observed an increase in pUb level during brain aging, further highlighting its relevance in this particular physiological process. To ensure reliable quantification of PINK1 and pUb levels, we used *pink1*^{-/-} mice and HEK293 cells as negative controls. For example, PINK1 levels were extremely low in control cells but increased dramatically after 2 hours of oxygen-glucose deprivation (OGD) and 6 hours of reperfusion (Figure 1H). Together, these

controls validate that the observed elevations in PINK1 and pUb are specific and linked to pathological or certain physiological conditions.

(2) Overexpression as a model for pathological conditions

To investigate whether the inhibitory effects of sPINK1* on the ubiquitin-proteasome system (UPS) are dependent on its kinase activity, we utilized a kinase-dead version of sPINK1* as a negative control. Since PINK1 has multiple substrates, we further explored whether its effects on UPS inhibition were mediated specifically by ubiquitin phosphorylation. For this, we used Ub/S65A (a phospho-null mutant) to antagonize Ub phosphorylation by sPINK1*, and Ub/S65E (a phospho-mimetic mutant) to mimic phosphorylated Ub. These well-defined controls ensured the robustness of our conclusions.

While overexpression does not perfectly replicate physiological conditions, it serves as a valuable model for studying pathological scenarios such as neurodegeneration and brain aging, where pUb levels are known to increase. For example, we observed a 30.4% increase in pUb levels in aged mouse brains compared to young brains (Figure 1F). Similarly, in our sPINK1* overexpression model, pUb levels increased by 43.8% and 59.9% at 30- and 70-days post-transfection, respectively, compared to controls (Figures 5A and 5C). Notably, co-expression of sPINK1* with Ub/S65A almost entirely prevented sPINK1* accumulation (Figure 5B), indicating that an active UPS can efficiently degrade sPINK1*. Collectively, these findings show that sPINK1* accumulation inhibits UPS activity, a defect that can be rescued by the phospho-null Ub mutant. Thus, this overexpression model closely mimics pathological conditions and offers valuable insights into pUb-mediated proteasomal dysfunction.

(3) Organization of the manuscript

We believe the structure of the manuscript is justified and systematically addresses the key aspects of the study in a logic flow:

- (a) Evidence for the increase of PINK1 and pUb in multiple pathological and physiological conditions.
- (b) Identification of the sources and consequences of sPINK1 and pUb elevation.
- (c) Mechanistic insights into how pUb inhibits UPS-mediated degradation.
- (d) Validation of these findings using *pink1*^{-/-} mice and cells.
- (e) Evidence of the reciprocal relationship between proteasomal inhibition and pUb elevation, culminating in neurodegeneration.
- (f) Demonstration of elevated pUb levels and protein aggregation in the hippocampus following sPINK1* overexpression, supported by proteomic analyses, behavioral tests, Western blotting, and Golgi staining.

Thus, this organization provides a clear and cohesive narrative, culminating in the demonstration that sPINK1* overexpression induces hippocampal neuron degeneration.

(4) Revisions to writing, referencing, and methodology

We will improve the clarity and flow of the manuscript, add more references to properly acknowledge prior work, and incorporate additional details into the Methods section to enhance readability and reproducibility. These improvements should address the organizational and technical concerns raised, while strengthen the overall quality of the manuscript.

Reviewer #3 (Public review):

Summary:

This study aims to explore the role of phosphorylated ubiquitin (pUb) in proteostasis and its impact on neurodegeneration. By employing a combination of molecular, cellular, and in vivo approaches, the authors demonstrate that elevated pUb levels contribute to both protective and neurotoxic effects, depending on the context. The research integrates proteasomal inhibition, mitochondrial dysfunction, and protein aggregation, providing new insights into the pathology of neurodegenerative diseases.

Strengths:

- *The integration of proteomics, molecular biology, and animal models provides comprehensive insights.*
- *The use of phospho-null and phospho-mimetic ubiquitin mutants elegantly demonstrates the dual effects of pUb.*
- *Data on behavioral changes and cognitive impairments establish a clear link between cellular mechanisms and functional outcomes.*

Weaknesses:

- *While the study discusses the reciprocal relationship between proteasomal inhibition and pUb elevation, causality remains partially inferred.*

The reciprocal cycle between proteasomal inhibition and pUb elevation can be initiated by various factors that impair proteasomal activity. These factors include A β accumulation, ATP depletion, reduced expression of proteasome components, and covalent modifications of proteasomal subunits—all well-established contributors to the progressive decline in proteasome function. Once initiated, this cycle would become self-perpetuating, with the accumulation of sPINK1 and pUb driving a feedback loop of deteriorating proteasomal activity.

In the current study, this reciprocal relationship between sPINK1/pUb elevation and proteasomal dysfunction is depicted in Figure 4A. Our results demonstrate that increased sPINK1 or PINK1 levels, such as through overexpression, can initiate this cycle. Crucially, co-expression of Ub/S65A effectively rescues the cells from this cycle, highlighting the pivotal role of pUb in driving proteasomal inhibition and establishing causality in this relationship. At the animal level, *pink1* knockout could prevent protein aggregation upon aging and cerebral ischemia (Figures 1E and 1G).

Mitochondrial injury is a likely source of elevated PINK1 and pUb levels. A recent study showed that efficient mitophagy is necessary to prevent pUb accumulation (bioRxiv 2023.02.14.528378), suggesting that mitochondrial damage can trigger this cycle. In another study (bioRxiv 2024.07.03.601901), the authors found that mitochondrial damage could enhance PINK1 transcription, further increasing cytoplasmic PINK1 levels and exacerbating the cycle.

- *The role of alternative pathways, such as autophagy, in compensating for proteasomal dysfunction is underexplored.*

Elevated sPINK1 has been reported to enhance autophagy (*Autophagy* 2016, 12: 632-647), potentially compensating for the impaired UPS. One mechanism involves the phosphorylation of p62 by sPINK1, which enhances autophagy activity. In our study, we did

observe increased autophagic activity upon sPINK1* overexpression, as shown in Figure 2I (middle panel, without BALA). This increased autophagy may help degrade ubiquitinated proteins induced by puromycin, partially compensating for the proteasomal dysfunction.

This compensation might explain why protein aggregation only increased slightly, though statistically significant, at 70 days post sPINK1* transfection (Figure 5F). Additionally, we observed a slight, though statistically insignificant, increase in LC3II levels in the hippocampus of mouse brains at 70 days post sPINK1* transfection (Figure 5—figure supplement 6), further supporting the notion of autophagy activation.

However, while autophagy may provide some compensation, its effect is likely limited. Autophagy and UPS differ significantly in their roles and mechanisms of degradation. Autophagy is a bulk degradation pathway that is generally non-selective, targeting long-lived proteins, damaged organelles, and intracellular pathogens. In contrast, the UPS is highly selective, primarily degrading short-lived regulatory proteins, misfolded proteins, and proteins tagged for degradation.

Together, we found that sPINK1* overexpression enhanced autophagy-mediated protein degradation while simultaneously impairing UPS-mediated degradation. This suggests that while autophagy may provide partial compensation for proteasomal dysfunction, it is not sufficient to fully counterbalance the selective degradation functions of the UPS.

- The immunofluorescence images in Figure 1A-D lack clarity and transparency. It is not clear whether the images represent human brain tissue, mouse brain tissue, or cultured cells. Additionally, the DAPI staining is not well-defined, making it difficult to discern cell nuclei or staging. To address these issues, lower-magnification images that clearly show the brain region should be provided, along with improved DAPI staining for better visualization. Furthermore, the Results section and Figure legends should explicitly indicate which brain region is being presented. These concerns raise questions about the reliability of the reported pUb levels in AD, which is a critical aspect of the study's findings.

We will include low-magnification images in the supplementary figures of the revised manuscript to provide a broader context for the immunofluorescence data presented in Figure 1. DAPI staining at higher magnifications will also be provided to improve visualization of cell nuclei and overall tissue structure. Additionally, we will indicate the brain regions examined in the corresponding figure legends, and incorporate more details in the Results section to provide clearer descriptions of the samples and brain regions analyzed.

The human brain samples presented in Figure 1 are from the cingulate gyrus region of Alzheimer's disease (AD) patients. Our analysis revealed that PINK1 is primarily localized within cell bodies, while pUb is more abundant around Aβ plaques, likely in nerve terminals. These additional clarifications and supplementary figures should provide greater transparency and improve the reliability of our findings.

- Figure 4B should also indicate which brain region is being presented.

The images were taken for layer III-IV in the neocortex of mouse brains, which information will be incorporated in the figure legend of the revised manuscript.

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